

SURVEY ARTICLE

A Decade of Inverse Kinematics Methods for Serial Manipulators: A Systematic Review

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ABSTRACT

Serial robotic manipulators find applications across diverse sectors such as industries, military, medical, space exploration, and underwater operations, where precision, efficiency, and safety are paramount. However, achieving effective control performance necessitates the effective resolution of the inverse kinematics (IK) problem inherent to manipulators. This challenge involves determining the requisite joint configurations to attain a desired endpoint position and orientation within the manipulator's workspace. The complexity arises due to the nonlinear equations and geometric relationships between cartesian space and joint space. Although various methods have been employed to address the IK problem in serial manipulators, there is a noticeable scarcity of publications comprehensively reviewing these techniques. Therefore, this paper undertakes an extensive review, thoroughly examining diverse IK methods applied in serial manipulators. This investigation of IK compiles literature spanning the last decade (2014–2023). Articles were collected from several web-based repositories such as WoS, IEEE Xplore, ACM Digital Library, Google Scholar, and ScienceDirect, and a preferred reporting items for systematic reviews and meta-analyses (PRISMA) protocol was adopted to acquire appropriate literature. The protocol involves examining the literature methodologies, major findings, and qualitative analysis. Thus, nearly 321 articles were screened and utilized in this review. Further, the IK methods are segregated into model-driven methods (analytical, geometric, algebraic, numerical, and other model-driven methods) and model-free/data-driven methods (metaheuristic (particle swarm optimization, genetic algorithm, and other metaheuristic methods), machine learning (artificial neural network, neuro-fuzzy, and reinforcement learning), and other data-driven methods). This paper offers a detailed exploration of limitations, challenges, performance comparisons, and future research directions of IK techniques. By conducting this study, we aim to assist prospective stakeholders in expanding their understanding of the remarkable potential of IK methods for serial manipulators.

1 | Introduction

A robotic manipulator is a mechanical device crafted to execute precise movements and manipulate objects across different environments. Typically, it consists of joints, links, and end-effectors,

exhibiting greater flexibility and capacity for handling substantial loads (Yen et al. 2019). Governed by programmable computer systems, they enable automated functionalities, thereby improving efficiency and safety across a range of tasks. In recent decades, robotics has garnered increasing attention from researchers due to

its widespread applications in scientific research and engineering. It finds utility in diverse fields, including space exploration, underwater surveying, military sectors, industrial (welding, painting, assembly, etc.), and medical processes (Dzedzickis et al. 2021). Significant contributions have been made to robotics, leading to the development and exploration of various types of robot manipulators. These include serial, cable-driven, parallel, mobile, and hybrid manipulators (Abbas et al. 2023; Antonov and Fomin 2023), as shown in Figure 1.

A serial manipulator is commonly constructed with a sequence of links interconnected by motorized joints, extending from a stationary base to an end-effector. Its functionality is versatile and has been used in automotive production lines, welding, paint jobs, space, and so on (Wu et al. 2022). In contrast, a parallel manipulator is a mechanical mechanism frequently utilizing multiple serial chains to uphold a solitary platform or end-effector. Specifically, these manipulators are implemented in medical, flight simulators, and deep sea exploration (L. Jin et al. 2018). On the other hand, cable-driven manipulators (as depicted in Figure 1c have flexible links (rather than rigid links) are driven by motors and pulleys, and find their application across various domains, including the construction of exoskeletons to assist individuals with disabilities, live broadcasting, and more. In the realm of mobile manipulators, robotic arms are affixed to a movable base, allowing them to explore broader workspaces. From an application point-of-view, mobile

manipulators are used for exploration in hazardous environments, handling explosives, and conducting railway track inspections (Rahman et al. 2023). Furthermore, hybrid manipulators (mostly a combination of series and parallel manipulators) are more advantageous with a larger workspace and greater stiffness than their counterparts, and their application is mostly found in surgical robots (Elsamanty et al. 2023). Out of these manipulators, serial manipulators have major participation in industrial applications.

However, understanding the kinematics of serial manipulators is essential to establish controllers and successfully deploy it in a precision-demanding application. The manipulator kinematics can be categorized into two types: forward kinematics (FK) and inverse kinematics (IK) (Singh and Banga 2022), as displayed in Figure 2. The FK involves determining the position and orientation of the end-effector linked to the robotic manipulator based on specified joint angles. Traditionally associated with FK, Denavit–Hartenberg (DH) parameters offer a systematic framework for establishing kinematic equations of the serial manipulator (Ayob et al. 2014; Bouzid et al. 2023a). On the flip side, the IK seeks to determine appropriate configurations of the manipulator's joints to achieve a desired pose in cartesian space coordinates (Wu et al. 2021).

In real-world scenarios, the orientation and position of the end-effector are often predefined, underscoring the significance of

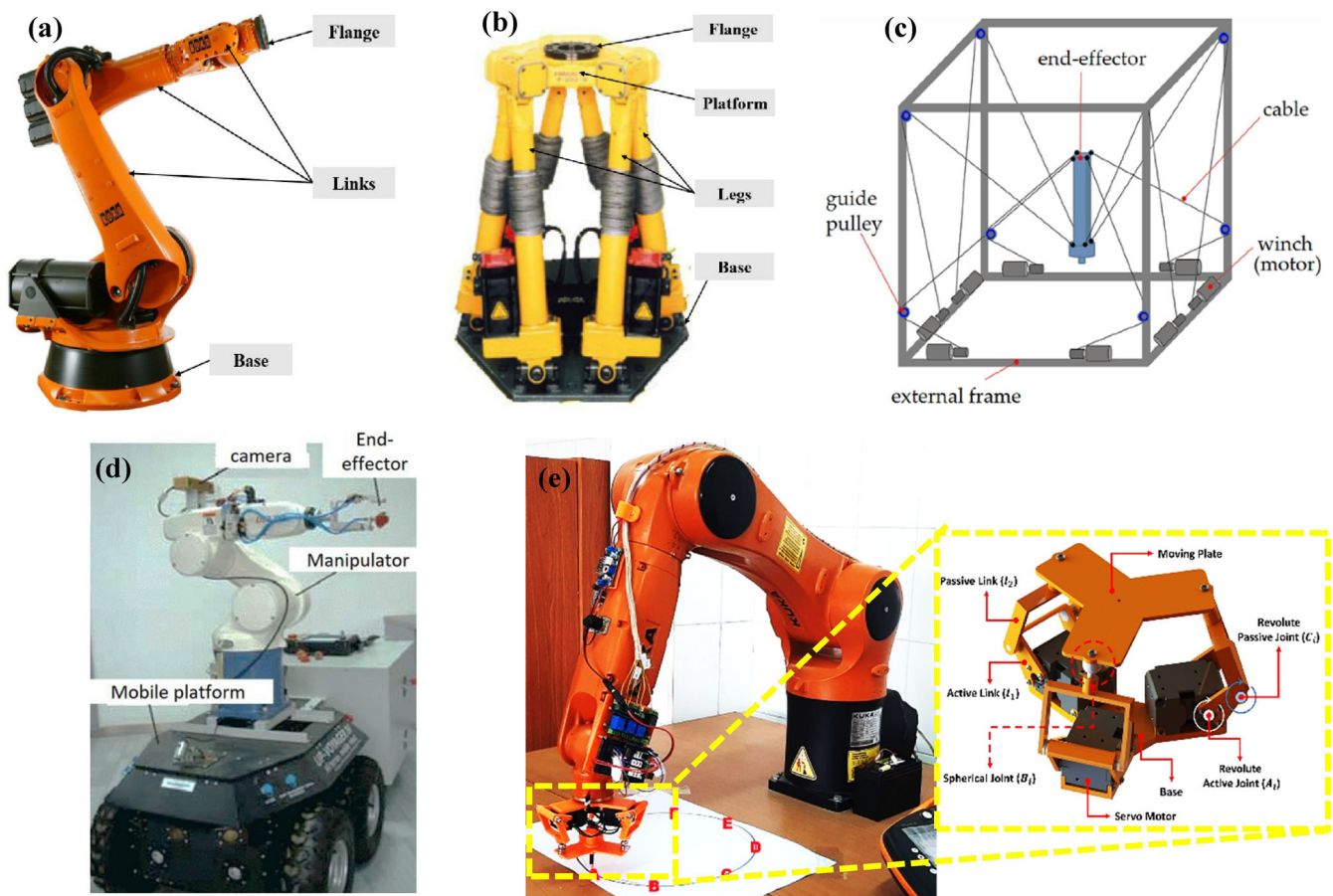


FIGURE 1 | Industrial manipulators: (a) serial (Wu et al. 2022), (b) parallel (Wu et al. 2022), (c) cable-driven (Jung 2020), (d) mobile (Rahman et al. 2023), and (e) hybrid manipulator (Elsamanty et al. 2023). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

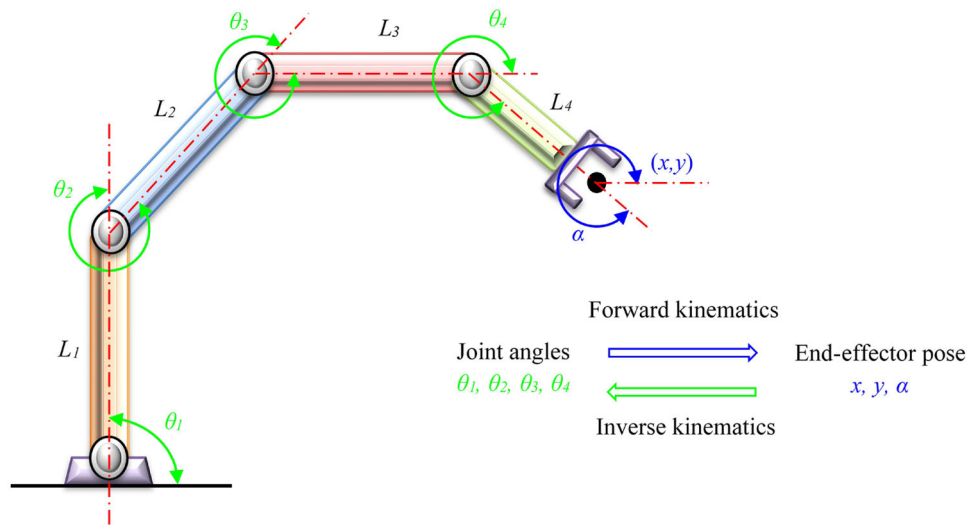


FIGURE 2 | Kinematics illustration of a serial manipulator. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70014)]

IK in enabling real-time trajectory planning and control of manipulators. Inverse kinematic solution (IKS) for multiple degrees of freedom (DoF) serial robotic manipulators involves complex nonlinear equations, demanding significant computation and time (X. Zhang et al. 2023). Consequently, researchers are committed to efficiently solving the inverse kinematic problem (IKP). Generally, the IK methods are classified as model-driven and model-free/data-driven methods. Wherein model-driven methods rely on the mathematical model and these methods encompass analytical (Tian et al. 2021), iterative (Lu et al. 2017), algebraic (X. Liu et al. 2019), and geometric methods (Xiao et al. 2021). Subsequently, each of these methods has its drawback and to resolve it, a combination (hybrid) of these model-driven methods was deployed to solve the IKP. Yet, formulating an algebraic expression for the robot's IK presents considerable challenges, particularly for redundant robots, where achieving stability in discrete time holds a significant hurdle. To overcome these, model-free/data-driven methods are opted since they are mostly data-driven approaches (no model is required). Moreover, data-driven methods can effectively solve intricate, time-consuming, and highly nonlinear problems, so they are actively utilized for solving the IK of industrial manipulators. The most preferred data-driven methods for solving IKP are: heuristic (particle swarm optimization (PSO; F. Liu et al. 2021), genetic algorithm (GA; Sui et al. 2015), and other heuristic methods (Firefly Algorithm (FA) (Hernandez-Barragan et al. 2018)), differential evolution (DE; Q. Zhang, Wang, et al. 2021), gray wolf optimization (GWO) (Cui et al. 2023), artificial bee colony (ABC) (Kumar et al. 2019, etc.), machine learning (ML) (artificial neural network (ANN; Šegota et al. 2021; Bouzid et al. 2023b) adaptive neuro-fuzzy inference system (ANFIS; Vladimirov and Koceski 2019), and Reinforcement Learning (RL; Kato et al. 2023)), and other data-driven methods (hybrid data-driven methods; P. Jha et al. 2015, etc.).

Despite significant advancements and a wide array of IK methods available, there is a notable scarcity of literature comprehensively reviewing the various techniques utilized in industrial manipulators. To the author's best knowledge, no comprehensive review has been conducted on the various IK

methods. Considering this motivation, the novelty and contribution of this paper are framed, and they are given below:

- Considering the critical role of IK in serial manipulators, this paper systematically examines a decade of advancements in IK methods. It evaluates various approaches, compares their performance, and identifies key innovations that have emerged in recent years.
- The preferred reporting items for systematic reviews and meta-analyses (PRISMA) methodology ensures a well-structured literature selection process. The bibliometric analysis identifies key trends in IK research, which are absent in textbooks and existing surveys.
- The IK methods are broadly classified as model-driven (numerical, analytical, algebraic, geometric, and other model-driven methods) and data-driven methods (metaheuristic (PSO, GA, and other metaheuristic methods), ML (ANN, ANFIS, and RL), and other data-driven methods).
- Additionally, the review explores the challenges and limitations of IK methods and outlines potential future research directions for advancing the field of IK of serial manipulators.

The subsequent sections of this paper are structured as follows: Section 2 outlines the methodology and systematic review protocol employed to gather relevant papers for this comprehensive review. Moving on to Section 3, it delves into the theoretical background of IK for manipulators. Section 4 explains the diverse methods utilized to solve the IKP in serial manipulators. Section 5 conducts an in-depth analysis of the methods explored, shedding light on their challenges and potential future scope. Finally, Section 6 concludes the paper with a summary of the key findings from this comprehensive review.

2 | Methodology and Systematic Review Protocol

A bibliometric and network analysis of IK methods was conducted using data from the Web of Science (WoS) core collection database from 2014 to 2023. The data was downloaded

(either in.txt or .RIS format) and uploaded in VOSviewer software (Van Eck and Waltman 2010). The VOSviewer software identifies thematic content based on keywords occurring more than two times in the uploaded file and 205 keywords were detected. Out of 205 keywords, the authors found that 35 keywords were more appropriate and these keywords were then used in a systematic review to outline research trends in IK methods for serial manipulators. Figure 3 illustrates the analysis, where node size indicates the frequency of recurrence and connections signify co-occurrence in articles. Notably, the node covering the widest area is “inverse kinematics problem” while “cuckoo optimization algorithm” appears smaller.

Further, the PRISMA technique is utilized for the identification of the relevant research works. Scholarly publications were gathered from web-based repositories across prominent international scientific databases: WoS, IEEE Xplore, ACM Digital Library, Google Scholar, and ScienceDirect. The search strings used were “inverse kinematics” AND “serial manipulator” OR

“analytical method” OR “geometric method” OR “numerical method” OR “algebraic method” OR “traditional methods” OR “particle swarm optimization” OR “genetic algorithm” OR “artificial neural network” OR “adaptive neuro-fuzzy inference system” OR “reinforcement learning” OR “hybrid methods” OR “heuristic methods” OR “machine learning”. The complete PRISMA process involved in collecting the suitable articles for this review is provided in Figure 4.

Thus, 321 articles were screened and finalized for this review paper. Additionally, Figure 5 visually represents the distribution percentage of each method within this review paper.

3 | Theoretical Background on Inverse Kinematics

Deeper knowledge of the kinematics of the manipulator is important to establish effective control strategies and as a result,

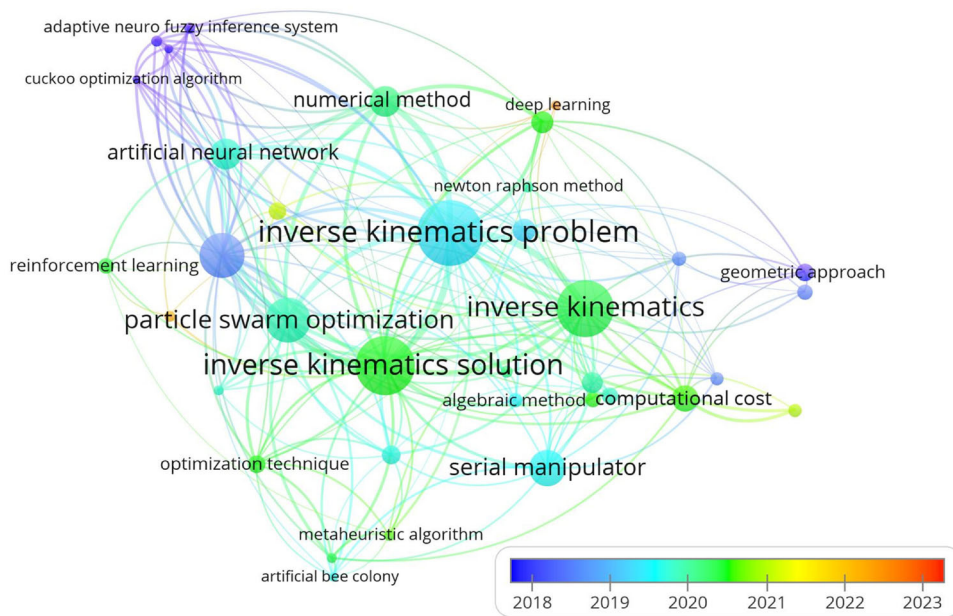


FIGURE 3 | Bibliometric analysis for the supplied keywords. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

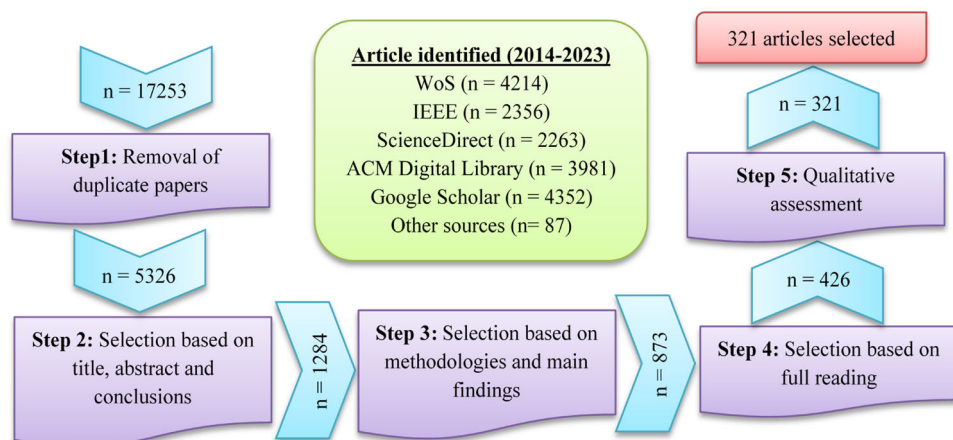


FIGURE 4 | PRISMA flowchart. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

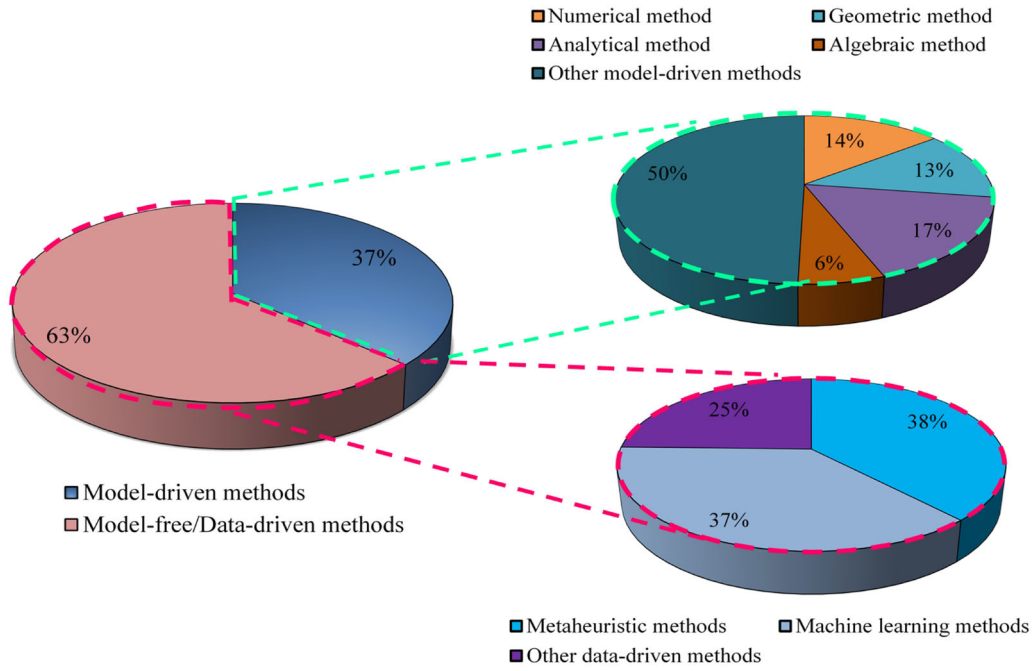


FIGURE 5 | Distribution of the different methods in this review paper. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70014)]

higher performance can be achieved. As discussed earlier, the frames/links of the robotic manipulator provide support for its effectors and joints that operate within an environment, and the motion is achieved through the elementary actions of joint effectors, which may include linear velocities or linear translations, angular velocities, and angular rotations. The generic vector $Q(q_1, q_2, \dots, q_n)$ is used to represent the position of joints, where q_i represents rotations or translations (generalized coordinates) and n defines the joints' quantities (emphasizes its dimensions). Generally, the position of the manipulator is determined by the FK (via DH), and $P(x_1, x_2, \dots, x_d)$ defines the generalized position, which is a vector quantity, wherein x_i represents the coordinate with respect to a specific frame along the i th dimension, and d is the dimension that relies on space depiction to execute motions.

3.1 | The Denavit–Hartenberg Forward Kinematics Modeling

Figure 6 illustrates an example of a serial manipulator delineated into a series of frames, beginning from base to end-effector, and where the different frames were fixed according to the DH convention. This approach facilitates a methodical and straightforward synthesis of the kinematics of the manipulator. From the two consecutive frames (i) and ($i + 1$), their relative displacements are deconstructed into series of elementary displacements, and each is denoted as a homogeneous transformation.

The resultant homogeneous transformation matrix is then expressed as given in Equation (1) (Rokbani, Mirjalili, et al. 2022):

$$T_{i-1}^i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

where θ_i is the joint angle and it denotes the relative angular position of the axis x_{i-1} toward the axis x_i around z_{i-1} , α_i (twist angle) is the angular rotation between the axes z_i and z_{i+1} , and around x_i , a_i is the distance of the frame's origin O_{i-1} to that O_i along the x_i axis, which stands for the normal distance of the axis z_{i-1} toward z_i along x_i , and d_i is the joint offset distance, also known as joint spacing, it is the relative position along the link z_{i-1} axis to the point where it cuts the x_i axis.

Equation (2) represents the transformation matrix constructed from frame 1 relative to frame 0 (the base), considering the parameters depicted in Figure 6:

$$T_0^1 = \begin{bmatrix} \cos \theta_1 & 0 & \sin \theta_1 & a_1 \cos \theta_1 \\ \sin \theta_1 & 0 & -\cos \theta_1 & a_1 \sin \theta_1 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

The decomposition of n elementary frames is crucial for the forward geometric model. It necessitates n transformations, in which every joint will have one transformation originating from the reference frame (x_0, y_0, z_0) to the end-effector frame (x_n, y_n, z_n). Equation (3) expresses resultant model arrived at from the product of each homogeneous transformation:

$$T_0^n = T_0^1 \times T_1^2 \times T_2^3 \times \dots \times T_{n-1}^n \quad (3)$$

The position (end-effector) arrived from the last column of matrix T_0^n , as follows:

$$P = \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} = T_0^n \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (4)$$

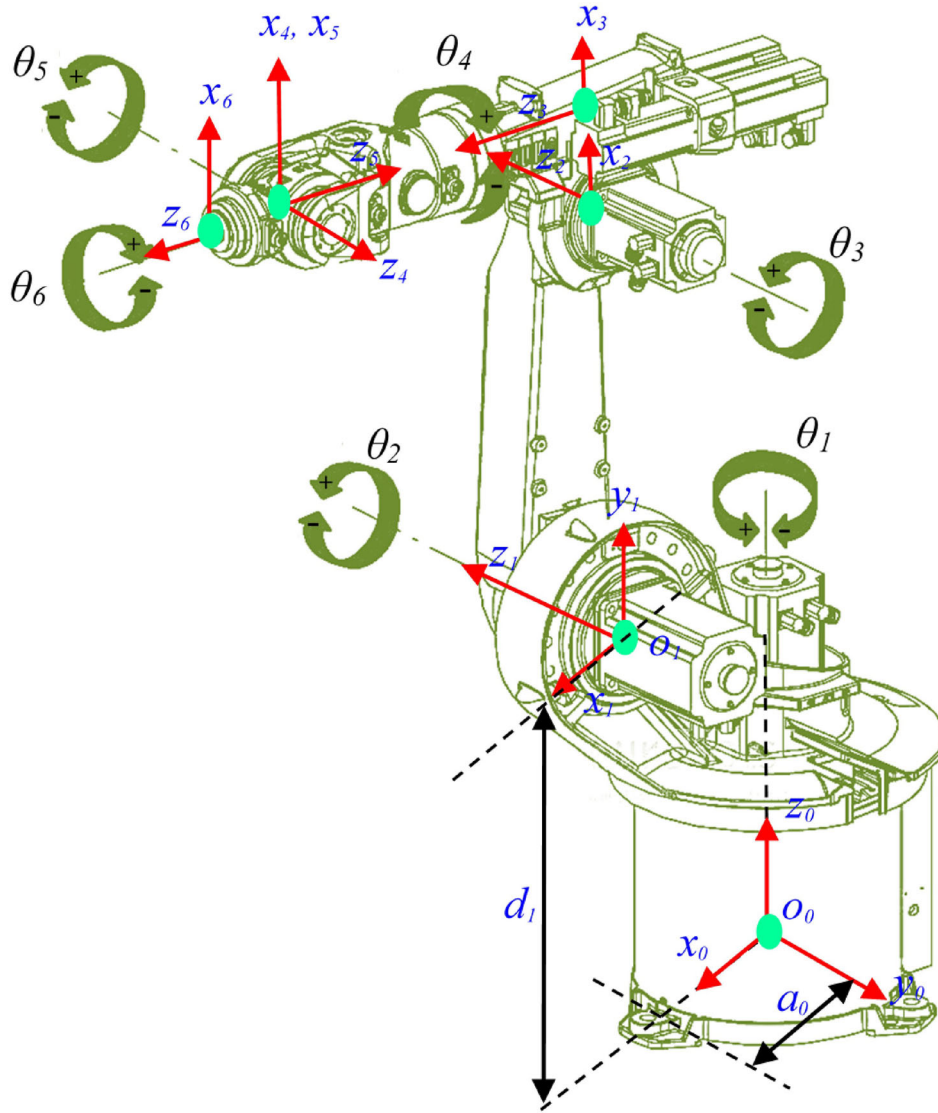


FIGURE 6 | Visualization of the relative transformations in DH modeling. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70014)]

where P is the FK solution, p_x represents the coordinate of the end-effector in the x -axis, p_y denotes the end-effector coordinate along the y -axis, and p_z is the coordinate along the z -axis.

3.2 | Inverse Kinematics Problem Statement

The relationship between joint position and frame position can be articulated through Equation (5). Through this equation, the system coordinates $P(x_1, x_2, \dots, x_d)$ within the dimension d can be calculated by providing a set of n joint positions $Q(q_1, q_2, \dots, q_n)$:

$$P^d = f(Q) \quad (5)$$

Here, $f()$ denotes the FK model, irrespective of the technique employed to derive it, and Q represents the joint position vector, encompassing translations and rotations.

The IK model facilitates the calculation of the necessary joint vector $Q()$ corresponding to a specified $P()$ -frame position.

In industrial applications where robots are utilized for manipulation and handling tasks, planning a path or motion typically involves a set of known points representing the objects to manipulate and assemble within a specified configuration frame. Thus, the IK model becomes crucial as expressed in Equation (6):

$$Q() = f^{-1}(P^d) \quad (6)$$

An IKS must incorporate the mechanical configuration of the systems, which could impose limitations or constraints on joint positions. Consequently, Equation (6) may encounter singularities or even lack a clear solution. Additionally, it should incorporate a collision avoidance strategy to prevent any potential damage from unintended contact among the workspace and robot. In the direction of circumventing the need for directly expressing the IK function $f^{-1}()$, the problem statement can be reformulated as a constrained optimization task using a heuristic approach, as outlined in Equation (7).

Find Q that meets the following conditions:

$$\begin{cases} Q \in D_q \\ P = f(Q) \\ \|P - P^d\| \leq \varepsilon_0 \\ P \in D_x \end{cases} \quad (7)$$

where D_q and D_x are the domain of acceptable joint positions and positions of frame, respectively, and ε_0 is the acceptable error.

4 | Inverse Kinematics Methods

Several techniques deployed to resolve the IK of serial manipulators are discussed in this section. These methods include model-driven (numerical, analytical, algebraic, geometric, and other model-driven methods) and model-free/data-driven methods (metaheuristic [PSO, GA, and other metaheuristic methods], ML [ANN, ANFIS, and RL], and other data-driven methods).

4.1 | Model-Driven Methods

4.1.1 | Numerical Method

In solving the IK of serial manipulators, numerical methods utilize iterative strategies to approach solutions. This technique aims to minimize the disparity between the desired and computed end-effector poses through successive iterations. Figure 7 illustrates a procedure of the numerical IK method. The procedure is repeated until optimal joint angle (less error) is reached.

A revolutionary numerical algorithm has been proposed by J. Li et al. (2021) to address the IKP. The manipulator was considered to have 6-DoF with nonspherical wrists. Initially, the nonspherical wrist was simplified into spherical. Through this simplified approach, the actual robot's end pose has been computed and position error (PE), that is, difference between target and achieved end pose is calculated. Subsequently, the IKS gradually converges towards the desired solution. Comparisons with the Newton-Raphson (NR) technique highlight the significance of the recommended method. Similarly, Yang et al. (2023) developed a comprehensive numerical methodology by simplifying the model. Here, a 7-DoF offset manipulator was considered. The PE and orientation error (OE) was estimated (via simplified model) and IKS (actual robot) was arrived. Xie, Sun, Wang, et al. (2022) introduce a speedup technique for numerical IK solvers to reduce execution time, enhancing performance. Results underscore effectiveness of the suggested method over existing alternatives. Further, Song et al. (2021) employed a novel numerical sequence processing method within a closed-loop framework to tackle the IKP of a 7-DoF redundant manipulator. Both simulation and experimental outcomes show the high performance of this technique.

Kumar et al. (2015) proposed a multi-dimensional numerical approach to solve the nonlinear IK equations, providing solutions for joint variables within the specified tolerance for end-effector accuracy. This includes addressing a planar 3-DoF

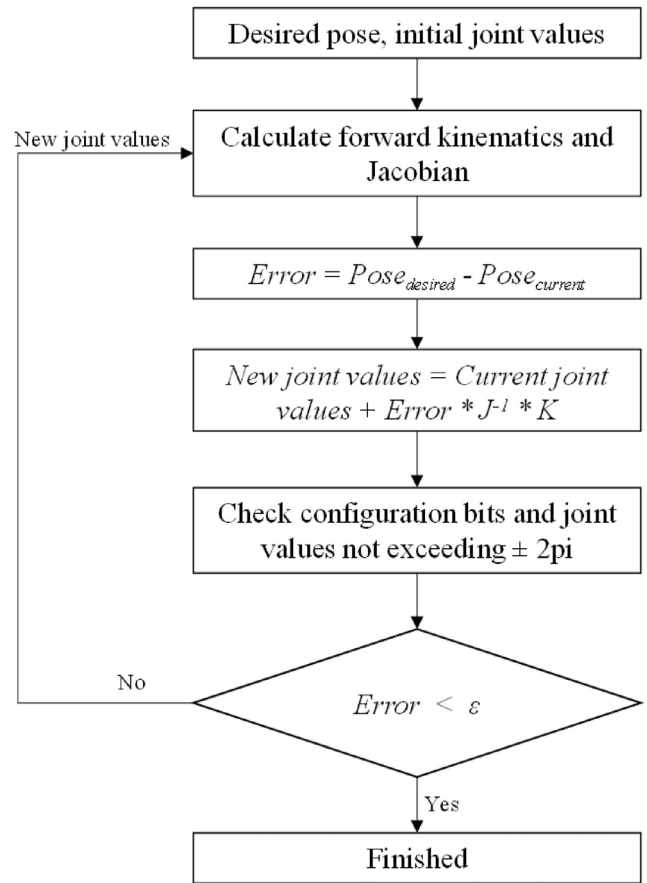


FIGURE 7 | Flowchart of numerical IKS (Bjoerlykhaug 2018).

serial link redundant manipulator as an illustrative problem. The numerical approach (Karabanov et al. 2022) chosen for solving systems of linear equations is the Gauss method, and its software implementation is discussed. During the software development process, object-oriented programming tools have been utilized. Verification of the software has been conducted using a 3D model of an actual industrial robot, demonstrating the effective performance of the IK technique. In Lloyd et al. (2022), a novel Quick Inverse Kinematics (QuIK), which utilizes first and second-order derivative information, has been presented. Results indicate that the QuIK is faster and more robust when compared to the NR algorithm. Sekiguchi and Takesue (2020) presented a new numerical approach for IK involving prioritized multiple targets. This method relies on joint-based damping and virtual spring model. Simulation validation using a 9-DoF robot reveals that the IKS is robust and faster than its counterparts. The IKS for a 7-DoF manipulator has been suggested by Cao et al. (2018) through a numerical algorithm utilizing an augmented Jacobian. By examining Jacobian matrix rank, the Jacobian singularities were analyzed. The proposed approach has been validated using the YASKAWA VA1400II manipulator. Demby's (2020) compared numerical and data-driven methods across various manipulators, ranging from 3-DoF to 7-DoF, for solving IKP. It has been observed that numerical solutions have proven to be easier to implement, faster, and more accurate than many data-driven methods. Moreover, experimentation conducted by Drexler (2016) has indicated that the numerical technique for the IKP converges more rapidly to the actual solution when employing the

Crank-Nicolson algorithm as opposed to the explicit Euler algorithm. Results reveal that the Crank-Nicolson-based IKS is preferable for solving the numerical-based IK of industrial manipulators.

Furthermore, J. Xu et al. (2018) developed a Hierarchical Iterative Inverse Kinematic Algorithm (HIIKA) for the IKS of a 6-DoF manipulator. The HIIKA has employed a two-level iteration algorithm, indicating better convergence, accuracy, and computational efficiency. Lu et al. (2017) suggested an iterative approach (falling under the numerical category) to solve the IK of 7-DoF robot. The plan point's current position, orientation, and arm angle are employed to compute the equivalent values for the succeeding plan point. The calculation yields the IKS, characterized by consistent arm angles and orientation matrices throughout the plan points. Effective results were observed. In L. Zhao, Zhao, et al. (2018), a novel iterative method has been unveiled for addressing the IKP of redundant manipulators. This solution utilizes iteration to generate valid configurations and implements a rank-based selection strategy for choosing an IKS. The technique's efficacy was assessed on manipulators with varying DoF. Fang et al. (2014) presented a generalized iterative technique for solving the IKP of redundant manipulators. Experimental findings highlight the efficacy of the suggested method in addressing the nonuniqueness problem associated with IK of redundant manipulators. In Fan et al. (2021), an iterative approach employing a multiple solutions strategy has been employed to tackle the IKP of a manipulator. The recommended method proves its efficacy through numerical examples and showcases improved computational time through parallel computing. H. Wang et al. (2015) attempted an iterative-based IKS for a 6-DoF manipulator by employing a dual quaternion scheme. This algorithm streamlines the solution process by assuming the preliminary value for the iterative variable, thus tackling challenges encountered while solving position and orientation equations using a DH-based solution. Simulation results for 100 end-frame poses reveal an average IK calculation time of 7.2 ms and typically 2–4 cycles for convergence, indicating rapid convergence.

4.1.2 | Geometric Method

The geometric method of solving IK of serial manipulators relies on DH parameters and homogeneous transformation matrices to represent the robotic arm's geometry. It entails specifying the desired end-effector pose, followed by the inversion of the transformation matrix, the decomposition of the rotation matrix, and the application of trigonometric solutions to ascertain the joint angles. Figure 8 shows the geometric method-based IKS proposed by Anschöber et al. (2023). Here, a geometric relationship has been established between the robot base and end-effector.

For many years, researchers have successfully resolved the IKP of 6-DoF manipulators with general geometry. Nevertheless, there continues to be a requirement for an efficient approach to develop IK formulas specifically tailored for a 6-DoF manipulator with simple geometry. In response to this necessity, Xiao et al. (2021) suggested a geometric method of disconnection and reconnection, showcasing its effectiveness through three examples. The results indicate unaffected solutions for wrist joint

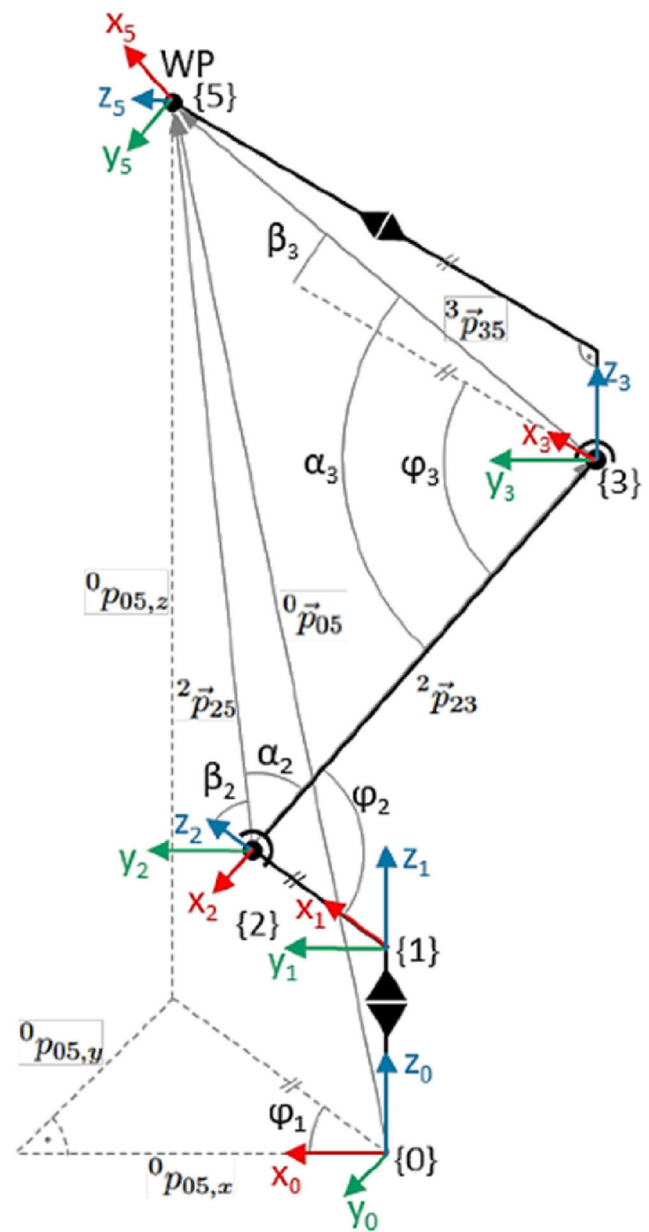


FIGURE 8 | Schematic of a serial robot arm representing geometric method-based IKS (Anschöber et al. 2023). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

variables even in the presence of singularities. Sinha and Chakraborty (2019) presented a geometric-based IKS for 7-DoF manipulators with joint offsets. Authors compute multiple IKS through a two-parameter search, leveraging the geometry of a redundant manipulator's structure. Demonstrations include achieving a minimum error bound in the end-effector position, even amidst random joint actuation and sensing uncertainties. Further, the IKP for the 5-DoF robotic manipulator is addressed with a geometric technique in Iliukhin et al. (2017). Implementation in the MATLAB environment showcases its significance through various simulation studies. In all the studies, the proposed algorithm surpasses other approaches in providing relatively better IKS. A novel geometric solution grounded in vector calculus to tackle the IKP has been given by Hadidi et al. (2023). The primary advantage of this technique lies in its

solutions, which manifest in a linear format with decoupled equations. Tested on a 6-DoF manipulator robot where positive outcomes were observed. The geometric-based IKS for a robot with an offset wrist is presented in Trinh et al. (2015). The desired positions are defined to match the solutions of four separate trigonometric equations involving the sixth joint angle. This method has been proven to be easily implementable and applied to various 6R serial robots.

Cai et al. (2021) conducted a comparison between geometric and algebraic approaches for solving the IKP. Experimental findings demonstrate that the end positions obtained using geometric and algebraic methods lie within the ranges of 2–4 and 6–14 mm, respectively. Petrenko et al. (2018) aimed to introduce a geometric solution for the IKP applicable to kinematic chains featuring rotational kinematic pairs. This geometric technique has played a pivotal role in achieving effective IKS, with input data comprising cartesian coordinates for the nodes within the chain. In Y. Liu, Wang, et al. (2015), a geometric-based IKS for the typical serial robot MOTOMAN with six joints has been executed. A simulation study corroborates the effectiveness of the proposed algorithm through a comparison of circles drawn before and after implementation, highlighting its significance. The concept of solving IK using a geometric approach has been introduced in Chavdarov and Naydenov (2022). This method is outlined for identifying regions where the robot can assume relative singularity configurations and a program is developed to operate within the AutoCAD environment to visualize results in two-dimensional and three-dimensional spaces. Zarrin et al. (2020) introduced a novel geometric approach for IK, incorporating the orientation and end-effector's location as inputs, resulting in a minimal computational error and a significant reduction in PE. He and Liu (2021) present the IKS for the 7-DoF FRANKA EMIKA PANDA robot with joint offsets, employing a geometric method. By examining the robot's geometry, the suggested solution unveils eight distinct joint configurations for a given end-effector pose and a fixed redundant parameter. Gliesche et al. (2021) developed a geometry-based IKS for a whole arm manipulator. A comparative analysis with two other approaches namely a commercial IK solver and a numerical method, has been conducted, affirming the superiority of the geometry-based IKS over its counterparts. H. Liu, Zhang, et al. (2015) presented a Pieper-criterion-based geometry method deployed in QJ-I robotic manipulator with 6-DoF to investigate the IK performance. Multiple case studies were conducted, and results revealed that the IKS was within acceptable limits.

4.1.3 | Analytical Method

The analytical approach to solving the IK of serial manipulators seeks closed-form solutions for joint angles by leveraging the geometric and kinematic characteristics of the robotic arm. These methods typically utilize mathematical techniques to explicitly articulate the correlation between the joint angles and the corresponding end-effector pose. Figure 9 shows the solution process of the analytical IK method for a 6-DoF manipulator suggested by Liao et al. (2020). This method has eliminated singularities and confirmed superior accuracy and efficiency compared to geometric-based IKS.

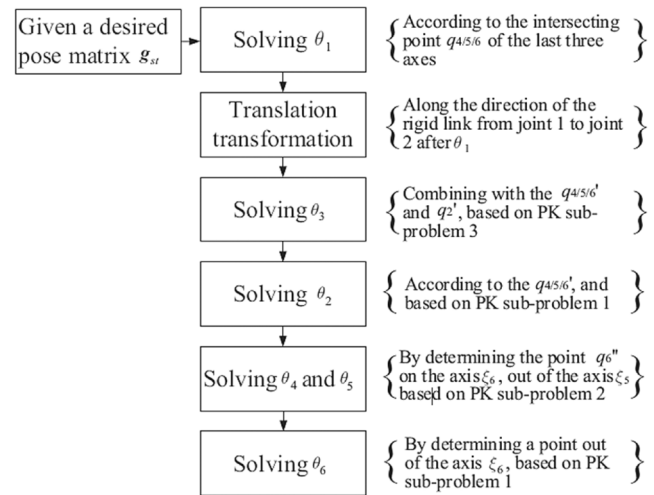


FIGURE 9 | Process of analytical IKS (Liao et al. 2020).

Tian et al. (2021) proposed an analytical method for the IK of a 7-DoF manipulator without offset. Simulation results demonstrate the advantages such as high precision, stability, and rapid calculation speed of the IKS. Oh et al. (2017) suggested an analytic IKS for a redundant 7-DoF manipulator, considering joint limits and self-collision avoidance. The suggested solution set, satisfying the subtask, is determined by stating the feasible arm angle range and its effectiveness is verified through simulation. Faria et al. (2018) introduced a novel analytical-based IKS for a 7-DoF robot, addressing joint limits and singularities. Two auxiliary parameters, global configuration and arm angle, are deployed to manage self-motion manifolds, with significant performance observed in simulation. Yan et al. (2014) proposed an analytical technique based on dual arm angle parameterization for solving the IKP, eliminating the algorithm singularity problem. Simulation results validate the recommended solution for both general and singularity cases. Ma et al. (2021) introduced a semi-analytical technique tailored for 7-DoF manipulators (with offset), leveraging arm angle parameterization to achieve high precision. The effectiveness of this algorithm has been validated through multiple simulation studies. Bouzgou et al. (2014) established an analytical approach for solving the IK of a 6-DoF manipulator (FANUC 200IC). Validation has been conducted through a simulation software platform that verifies IK results in a virtual reality environment based on MATLAB software integrated with the CAD model. In Singh et al. (2019), IKS for PUMA robot using an analytical method has been suggested. All the angle values are solved by DH, and the solution involves finding the homogeneous transformation matrix describing the tool frame's position and orientation with respect to the global reference frame. Effective IK results are observed. Garnayak et al. (2020) proposed an analytic-based IKS for a 6-DoF manipulator. After solving the IK equations, possible solutions are cross-checked using a virtual manipulator module and the IK results were satisfactory. Sun et al. (2017) developed an analytical-based IKS for a 6-DoF manipulator. The chosen robot was configured with 6-DoF in a 6 R arrangement. Utilizing the DH algorithm, the analytical approach is deduced, followed by an in-depth examination of the robot's IK. Finally, the performance of IK equations was tested via simulation studies. W. Xu et al. (2014) developed analytical and semi-analytical-based IKS for a 7-DoF robot considering joint lock failure. The

simulation study shows the significance of the developed approach giving the most suitable IKS through three configurations subject to joint lock failure.

In B. Wei et al. (2020), the AUBO-I5 manipulator undergoes kinematic modeling and analysis. The analytical method is applied to solve the IKP for this manipulator, with simulation studies conducted in the MATLAB environment. The suggested solution demonstrates significant improvements in the manipulator's IK performance. In H. An et al. (2014), an analytical approach for the 7-DoF robot arm with non-zero joint offsets is presented, offering two closed-form solutions: the shoulder-elbow-wrist IK and the θ_1 -based solution. Both address algorithmic singularities within the workspace, providing simple, fast, and accurate results according to simulation studies. Luo et al. (2014) proposed an analytical-based IKS for a 7-DoF redundant manipulator with offsets. The key component involves deriving a closed-form solution using different joints as redundancy based on a parameterization method. This gives various null space self-motion patterns for the arm, and the derived equations are validated with the chosen manipulator with offsets. W. Xu et al. (2016) introduced an analytical IKS based on dual arm angle parameterization. A notable contribution is the derivation of the absolute reference attitude matrix, which is essential for resolving analytical IK. Simulation results, including the algorithm's singularity condition, confirm the efficacy of the method. Gong et al. (2019) developed an IKS for a unique joint-configured 7-DoF manipulator. An analytical approach has been employed to explicitly determine the entire set of feasible IKS for redundant manipulators. The effectiveness and validity of these methods have been confirmed through kinematic simulations. This study (Brandstötter et al. 2014) validates an analytical solution for calculating the IK of standard serial manipulator arms with up to 6-DoF. The technique illustrates the analytical computation of all feasible joint angles corresponding to a given end-effector pose. Through this, a rapid and versatile algorithm has been developed, relying on a concise parameter set. Q. Chen et al. (2015) introduced an IKS leveraging screw theory. Additionally, a novel IK technique has been implemented, incorporating Paden-Kahan sub-problem and improved sub-problem algorithms. Using a 6-DoF QIANJIANG I serial robot, the effectiveness of this approach has been assessed.

4.1.4 | Algebraic Method

In the context of solving the IK of serial manipulators through the algebraic method, the process includes formulating a set of nonlinear equations that describe the relationships between joint angles and the desired end-effector pose. Typically, these equations are tackled using algebraic techniques such as symbolic algebra solvers for effective solutions.

In X. Liu et al. (2019), an IKS has been derived through the application of algebraic methods. The motion law for each joint variable is determined based on a trajectory planning technique for continuous paths. Subsequently, the correctness of the IK algorithm is validated through ADAMS simulations. F. Liu et al. (2017) presented an algebraic approach to tackle the IKP. The IKS for the kinematic equations of the robotic manipulator is

derived, primarily encompassing the four joint variables. Simulations conducted in ADAMS demonstrate the effectiveness of the suggested method. Moreover, the Conformal Geometric Algebra (CGA) method that falls under the algebraic IK method, as mentioned in El-Sherbiny et al. (2018a), has been extensively explored by researchers for solving the IKP. The CGA is a mathematical framework that combines geometric and algebraic representations. Kleppe and Egeland (2016) showcase the IKS using CGA to provide consistent solutions for joint angles in different configurations. Furthermore, the method creates a connection between the solution and the DH parameters of the robot. Successful implementation and extensive testing of the technique have been carried out across the entire workspace of the manipulators. In the context outlined by Tørdal et al. (2017), an industrial robot's IKS has been presented by employing CGA. The extraction of accurate joint angle signs is accomplished by utilizing multi-vector coefficients and applying FK. The effectiveness of this developed IKS has been established and benchmarked successfully using a 3-DoF manipulator. In Lechuga-Gutierrez et al. (2022), an approach has been devised utilizing screw rotors and CGA, resulting in efficient update rules to address the problem of inverse position kinematics of n -DoF kinematic chains with revolute joints. Simulations employing various DoF models, alongside experiments, prove the recommended IKS. In Zaplana et al. (2022), CGA-based IKS has been employed for a serial robot. Here, the problem has been partitioned into two sub-problems: orientation and position. The orientation issue is tackled by decomposing the rotor into three simpler rotors, while a geometric strategy has been developed for each joint combination to solve the position problem.

4.1.5 | Other Model-Driven Methods

The literature that established hybrid model-driven methods (combines two or more model-driven methods) and other model-driven methods to solve the IKP of serial manipulators are analyzed here.

Routray et al. (2019) seek a balance between the complexities of geometric and iterative methods to solve the IKP for a 4-DoF robotic arm. An iterative technique is employed for specific links, while geometry is applied to the rest, achieving the desired end location with minimal error within the work envelope constraints. A combination of geometric and algebraic approaches for IKS has been suggested by Huda et al. (2022). The hybrid model-driven-based IKS was successfully applied to a 6-DoF robot. Meanwhile, the average PE was noted to be 2.08 and 12.9 mm at the x and z coordinates, respectively. R. Zhao, Shi, et al. (2018) combined geometric, algebraic, and analytical techniques with Paden-Kahan sub-problem and matrix theory. The developed solution was implemented on a 6 R manipulator, yielding effective IKS with high accuracy. In Saadah et al. (2020), the IKP is tackled using geometric and analytical techniques. The developed algorithm was tested in a MATLAB environment considering the ABB IRB 4400 robot. Superior IK performance of the recommended approach was noted during simulation studies. Similarly, in Lin and Min (2015), a geometric-analytical solution is adopted for a 6-DoF manipulator's IK, validated on WUST ARM. Experimental validation

confirmed that the running time adequately meets the real-time control needs and also has shown relatively less error. Abdullah et al. (2019) focused on IKS for a robotic arm, providing algebraic-trigonometric nonlinear analytical models. This model has been developed from the extensive manipulation of algebraic arithmetic, Pythagorean identities, sum-and-difference, and trigonometric rules to acquire accessible joint configurations of a two-segmented manipulator. Y. Wei et al. (2014) tackled the spatial nR robot IKP using both semi-analytic and general techniques, overcoming numerical method limitations related to accuracy and introducing a real-time aspect, validated through testing with a special 6 R robot. In Zhou et al. (2022), an improved IKS has been suggested for 6-DoF robot manipulators (HSR-BR606 with an offset wrist). The method combines analytical reconstruction with numerical iteration, demonstrating higher accuracy and shorter calculation time than the model-driven IK technique in a comparative experiment. Ananthanarayanan and Ordóñez (2015) developed a hybrid model-driven IK method (analytical and numerical) to solve the IK of manipulators effectively. Multiple simulation studies (considering a 9-DoF robot) were conducted to test the effectiveness of the hybrid technique. Satisfactory results were observed and the method is applicable to provide IKS for the industrial manipulator. Similarly, a hybrid model-driven IKS by combining analytical and numerical approaches has been introduced by M. Jin et al. (2020). Further, a comparative study was conducted among the developed IKS, Closed-Loop Optimization (CLO), Jacobian Pseudo-Inverse (JPI), and ANN. On comparing CLO, JPI, and ANN, the proposed hybrid model-driven method improved the PE by 26.90%, 97.45%, and 78.56%, respectively. The research conducted in Manongsong and Capco (2023) offers a solution to the IKP for specialized six-joint manipulators utilizing the algebra of dual quaternions and techniques from Gröbner basis, linear algebra, and the elimination theory were utilized to solve IKP algebraically. To exemplify, an example of a special manipulator was provided and its IKP was solved. J. Wang, Shao, et al. (2018) suggested analytical and CGA-based IKS for a 6-DoF robotic manipulator. The joint angles (θ_1 , θ_5 , and θ_6) were solved using the analytical method, and the remaining joint angles (θ_2 , θ_3 , and θ_4) were solved by CGA. The IK performance was verified and validated in MATLAB and Rviz simulation environments.

Semwal et al. (2021) established forward and backward reaching inverse kinematics (FABRIK)-based IKS. Multiple simulation studies were conducted considering 3-DoF and 6-DoF manipulators and compared to a feed-forward neural network (FFNN). The findings indicate that the FABRIK algorithm appears to outperform the FFNN in providing more accurate IKS. In X. Zhang, Fan, et al. (2021), a fusion of the GPM, WLN, and damped least squares (DLS) has resulted in the creation of an Enhanced Weighted Gradient Projection Method (EWGPM) tailored for resolving the IKP in redundant manipulators. Through simulation trials conducted in two scenarios, the EWGPM has shown its higher efficacy compared to alternative methods. L. Zhao et al. (2021) have unveiled a novel algorithm combining Newton-based and first-order methods to produce collision-free IKS. Through simulations and physical experiments with a BAXTER robot in dynamic environments, the solution has proven superior in error and computational efficiency compared to existing techniques. J. Wang et al. (2019)

have merged a sampling-based planning technique with analytical IK computation, whereby configurations satisfying constraints are randomly sampled from the manifold and linked to motion paths in joint space. This algorithm efficacy has been showcased on Willow Garage's PR2 simulation platform, generating trajectories amidst a diverse array of orientation-constrained scenarios for robotic manipulation. An Improved Clamping Weighted Least-Norm Method (ICWLNLM) has been presented for the IKP of redundant manipulators with joint limits by J. Wan, Wu, et al. (2018). The recommended ICWLNLM adds an elastic field function into the standard approach to sustain the constraints of joint angular velocity limits. Case studies have revealed the effectiveness of the proposed method. A workspace-density-based approach to solve the IK of a discretely actuated ball-joint manipulator has been presented in Dong et al. (2015). Authors have exhibited the method with several numerical examples, providing a solution that approximately minimizes the deviation of the end-effector and its computational complexity. In Xie, Sun, Chen, et al. (2022), a hybrid method integrating symbolic pre-processing, Sylvester dialytic elimination, and eigen decomposition for IKS has been introduced. Its implementation on the BAXTER robot with 7-DoF revealed that the PE was about ± 0.5 mm and also average deviation was 7.942 mm. H. S. An et al. (2018) presented a generalized solution for an IK sub-problem using an exponential formula product, verified with a 6-DoF manipulator in a case study, offering applicability to various serial manipulators. A dual quaternions method has been studied by Dantam (2021) to solve the IKP of the manipulator. The IKS was developed considering the UR10 model and several simulation studies were conducted. Simulation results show that the recommended IKS yields a 300%–500% improvement in IK performance. X. Wang et al. (2020b) developed optimized Gaussian damped least squares (GDLS) and optimized elimination methods and deployed them in the MOTOMAN robotic manipulator (6-DoF) to test the IK performance. The optimized elimination method showed better accuracy and optimized GDLS had lesser computation time.

Trutman et al. (2022) deployed a globally optimal solution for the IKP concerning a standard serial 7-DoF manipulator, employing a polynomial objective function. This technique has been applied to the 7-DoF KUKA LBR IIWA manipulator, indicating its capability to compute the optimal IK or ascertain infeasibility in 99.9% of the examined poses. Faroni et al. (2020) introduced a novel approach to ensure that the solution of the IKP maintains deviation from the nominal joint-space trajectory below a predefined threshold. This is achieved through the implementation of linear constraints and the automatic adjustment of secondary task weights. Both numerical and experimental analyses have confirmed the efficacy of this technique. A new predictive task-scaling algorithm has been suggested in Faroni et al. (2019), allowing the solving of the IK of a redundant manipulator taking into account the kinematic limits of the robot. In this way, the computational complexity has been dramatically decreased, making the approach suitable for online implementation. In Zaplana and Basanez (2018), a method has been devised to streamline redundant manipulators into nonredundant configurations by carefully selecting specific joints, termed as redundant joints, and adjusting their joint variables accordingly. This selection process, facilitated by a

comprehensive workspace analysis, not only identifies redundant joints but also creates an upper boundary for the range of distinct closed-form solutions achievable for a given pose. Shirafuji and Ota (2019) have computed the trajectory realized by implementing joint displacements acquired through an optimization step using the differential IK approach. This approach derives joint velocities based on their relationship as represented by the Jacobian matrix to achieve the desired end-effector velocity. The differential IKS has been enhanced by incorporating an optimal linear combination of null space basis vectors from the corresponding velocity Jacobian in Reiter et al. (2016). The effectiveness of this method has been executed using an industrial robot with 7-DoF, with comparisons made against a joint space decomposition IK algorithm. Taking inspiration from the advantages of the weighted Least-Norm (WLN) and Gradient Projection Method (GPM), H. W. J. Wan, Yao, et al. (2018) has developed the Weighted Gradient Projection Method (WGPM) to address the IKP of the redundant manipulator. Simulations and experimental assessments have been conducted utilizing a 7-DoF redundant manipulator to authenticate the WGPM. Through comprehensive analysis, the suggested IK technique's superior performance has been confirmed when compared to alternative algorithms. Abdullah et al. (2022) investigated IKS of a two-segmented manipulator through Cramer's and trigonometry rules. The former section of models has been obtained using trigonometric identities to solve for the second joint angles, while the latter part has been developed using Cramer's rule to determine the first joint angle. The proposed method has been successfully validated. da Silva et al. (2021) analyzed two approaches for solving the IKP namely: Paul's method (uses DH matrix) and Gröbner Bases Theory (GBT). Each method has solved the IKP of a robot, with GBT proving more computationally efficient. Moreover, some of other model-driven methods utilized to solve the IKP are tabulated in Table 1.

4.2 | Model-Free/Data-Driven Methods

Model-free/data-driven methods, highlighted in Figure 10a for their significance (Narayanan and Ramakrishnan 2021; Lakshmi Narayanan and Ramakrishnan 2022), have garnered widespread application across diverse fields due to their remarkable advantages. This section conducts a thorough analysis of data-driven techniques applied to address the IKP in serial manipulators. Moreover, the general steps involved in the implementation of data-driven methods are provided in Figure 10b. Among these methods, the metaheuristic-based methods solve IK as a constrained optimization problem. Whereas, the ML methods use historical data to predict joint configurations and finally the other data-driven methods mostly discuss hybrid data-driven methods.

It is noteworthy that fuzzy logic has been utilized from the early stages for solving the IKP of serial manipulators (Kim et al. 1993; Schacherbauer and Xu 1993; Morishita and Tojo 2013; Borboni 2001; Shen et al. 2006). Despite its historical use, recent research has not shown significant developments in fuzzy logic-based IKS for serial manipulators. Thus, the fuzzy logic method is neglected in this section. Analysis of data-driven algorithms addressing IKP is as follows.

4.2.1 | Metaheuristic Methods

4.2.1.1 | Particle Swarm Optimization. The PSO evolutionary algorithm draws inspiration from the collective behavior observed in bird flocks (Danaci et al. 2023). Figure 11a shows the trajectory of a particle as it moves through a single step of the PSO method and Figure 11b illustrates the detailed PSO process. In PSO, particles explore the function's search space, aiming to minimize the cost or objective function related to achieving the desired manipulator position. The direction of each particle's movement is determined by a combination of its current position, its best-known position, and the best-known positions of its neighbors. This adjustment is based on both the individual particle's experience and the collective knowledge of the swarm. These steps are repeated until the manipulator's optimal joint configuration is found.

In F. Liu et al. (2021), a Parallel Learning Particle Swarm Optimizer (PLPSO) has been implemented for IKP in a robotic manipulator, exhibiting superior performance compared to three other heuristic methods. The PLPSO algorithm's practicability was verified through tests on a UR5 manipulator. In Sridhar Reddy et al. (2023), IKS for a 9-DoF manipulator (a welding robot) based on multi-objective PSO has been suggested. The method uses constrained multi-objective optimization and integrates obstacle avoidance techniques for collision prevention during welding. MATLAB simulations have confirmed the successful reaching of desired locations, emphasizing adaptability for various weld joint shapes. In Sancaktar et al. (2018), an adaptive particle swarm optimization (APSO) has been developed for the IKS of a 6-DoF robot. Unlike the classic PSO, the proposed technique prevents local particle jamming by repositioning and restarting the algorithm until a specified criterion is met. Validation of a 6-DoF robot establishes greater IK performance over the traditional PSO. Similarly, Deng and Xie (2021) suggested an APSO for IKP, which was tested on two serial robotic manipulators. The results show that the proposed fitness function with APSO efficiently and accurately solves IKP for multi-DoF manipulators, outperforming several PSO variants from diverse techniques. Likewise, in Ning et al. (2023), an APSO-based IKS has been suggested to improve the speed and precision of the 7-DoF manipulator. This technique utilizes a quality evaluation criterion and probability transfer mechanism, exhibiting effectiveness and reliability through simulations and prototype experiments on a Robot Operating System (ROS)-based platform. In Dereli and Köker (2020a), a quantum behaved particle swarm optimization (QPSO) has been applied for efficient IKS in a 7-DoF serial manipulator. It outperforms standard PSO, ABC, and FA in terms of computation time, iterations, and particle count. The study validates the algorithm's effectiveness through scenarios involving single and a hundred different positions, confirming its higher IK performance. Ayyıldız and Çetinkaya (2016) solved the IKP using PSO and QPSO and compared the IKS to GA via 4-DoF robot. Results reveal QPSO's effectiveness in achieving the IKS, outperforming other algorithms. In Yiyang et al. (2021), an improved particle swarm optimization (IPSO) approach for general robot IK has been proposed, featuring dynamic inertia weight adjustment and multiple populations. Tested on a CO-MAU NJ-220 robot, the IPSO demonstrates higher stability, improved convergence accuracy, and speed compared to

TABLE 1 | Various other model-driven methods deployed in IKP of serial manipulators.

Reference	Year	Model-driven methods	Manipulator	DoF	Validation Simulation/ Experiment	Inference
Ojha et al. (2023)	2023	Improved cyclic coordinate descent	DOBOT CR3	6	Experiment	Validation through experiments underscores the algorithm's robustness and practical applicability.
Ratajczak (2015)	2023	Extended Jacobian	PUMA	4	Simulation	The proposed approach was effective in providing IKS.
Yoshizawa et al. (2023)	2023	Quantifier Elimination based on GBT	—	3	Simulation and experiment	This method of solving the IKP checks if a solution exists by counting the real roots of polynomial equations. The suggested method was computationally efficient and validated via simulation and experiment.
Guzmán-Giménez et al. (2023)	2023	GBT	PUMA	3	Simulation	The proposed method yielded effective IKS, and the error values were within acceptable limits.
Nguyen and Marvel (2022)	2022	Analytical + statistical	UR5	6	Experiment	The standard deviation of OE was between 0.005 deg and 4.26 deg.
F. Chen and Ju (2022)	2022	Improved dixon elimination	—	6	Simulation	The PE and OE were noted to be around 10^{-15} .
Pikalov et al. (2022)	2022	Vector model-based IK	KUKA KR 6 R900	6	Experiment	The PE was found to be less than 0.005 mm.
Safeea et al. (2021)	2021	Modified DLS	KUKA IIWA	7	Experiment	The observed PE was noted to be ± 0.15 mm.
Kim et al. (2021)	2021	Null space projection and potential field	ROMAN 7	7	Experiment	This technique has successfully executed the required task within the target workspace and angular limitations in real-time.
Ricardo Xavier da Silva et al. (2021)	2021	GBT	KUKA	7	Simulation	The DH model has been established, and joint configurations were obtained using GBT. Validation confirmed the method's ease of implementation and accuracy.
Park et al. (2020)	2020	Jacobian + Modified potential field	—	7	Experiment	The proposed method can realize real-time calculations (obstacle avoidance) and effective IK results were noted.
Maric et al. (2020)	2020	Convex optimization	—	5 & 7	Simulation	The PE of the proposed approach was relatively lower than that of the FFNN.
Toz (2020)	2020	Chaos-based vortex search method	—	6	Experiment	The IKS of the developed technique was better than the standard vortex method.
Tringali and Cocuzza (2020)	2020	Global kinematic planner	—	3	Simulation	The noted OE was between 0.02 and 0.009 rad.

(Continues)

TABLE 1 | (Continued)

Reference	Year	Model-driven methods	Manipulator	DoF	Validation/ Simulation/ Experiment	Inference
X. Wang et al. (2020a)	2020	Optimized elimination, optimized GDLS	YASKAWA-EPX2050	6	Experiment	The recommended IKS was more effective than the conventional elimination and GDLS techniques.
Guzmán-Giménez et al. (2020)	2020	GBT	PUMA	3	Simulation	The suggested GBT relies on the geometry of the robot's structure. The error and computing time were significantly less than its counterparts.
Lu et al. (2015)	2019	Arm-wrist separateness method	TX90 XL	6	Experiment	The algorithm put forward has delivered an efficient IKS, presenting significant potential for enhancing industrial applications.
An and Lee (2019)	2019	Prioritized- based IK	KUKA LWR	7	Experiment	The suggested approach showed satisfactory IK performance.
Ren et al. (2019)	2019	FABRIK	DOBOT	4	Simulation	FABRIK showed more effective IKS than the model-driven technique.
Dai et al. (2019)	2019	Mixed-integer convex optimization	ABB IRB140	6	Experiment	The findings demonstrate the efficacy of the suggested approach in achieving better IKS.
Scherzinger et al. (2019)	2019	Improved IK method	UR10	10	Experiment	The IK results of the developed method were better than the inverse Jacobian solution.
Z. Chen et al. (2018)	2018	Vector product + DLS	KINOVA	7	Simulation	IKS was within the acceptable limits.
J. Zhao, Xu, et al. (2018)	2018	DLS + NR	—	7	Simulation	The results substantiate the importance of the suggested method in delivering IKS.
Y. Wan, Kou, et al. (2018)	2018	DLS	—	7	Experiment	Implementation of the IK approach was successful.
Blanchini et al. (2017)	2017	Convex programming problem	PA-10	7	Simulation	The algorithm involves treating the joints as point masses in a hypothetical gravity field, incorporating constraints to account for the link lengths, and framing the IK. Simulation results validate the effectiveness of IKS.
Di Vito et al. (2017)	2017	DLS	KINOVA JACO ²	7	Simulation	The performance of the DLS was under acceptable limits.
Pfurner (2016)	2016	Novel closed-form IKS	—	6	Simulation	This solution opened up a new opportunity to explore the IK of the manipulator.

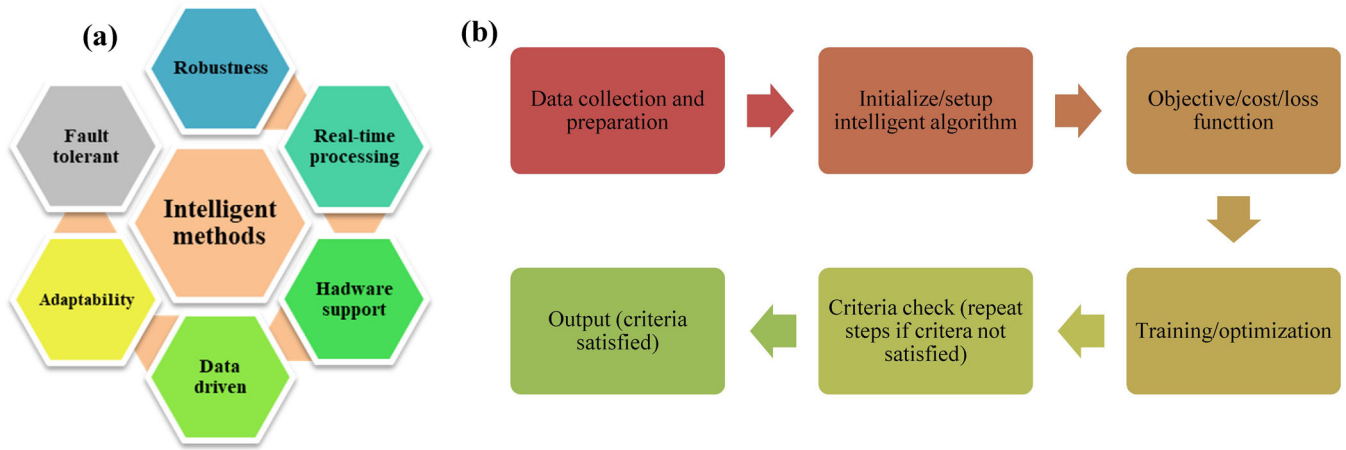


FIGURE 10 | Data-driven methods: (a) Significance and (b) general implementation steps. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70014)]

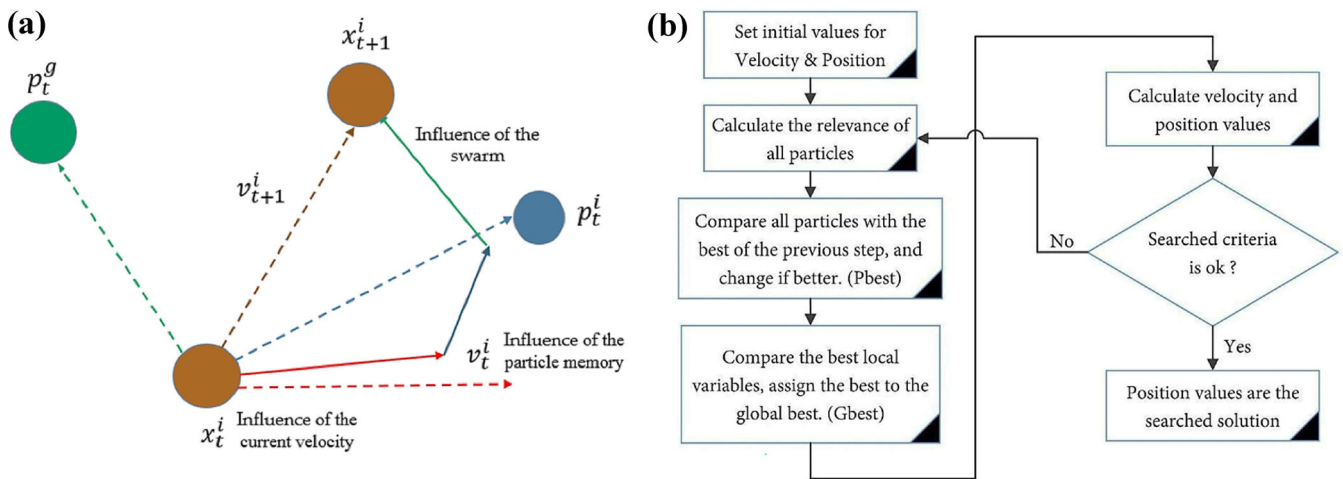


FIGURE 11 | Particle swarm optimization: (a) Schematic (Sharmin et al. 2023) and (b) process flow (Sridhar Reddy et al. 2023). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70014)]

conventional PSO and multi-subswarm method, offering an efficient IK technique. Similarly, J. An et al. (2020) presented an IPSO approach for solving IKP in robot manipulators without closed-form solutions. Incorporating the global optimal position of particles, the method exhibits enhanced performance by effectively solving 6-DoF manipulator IKP with offset wrists, outperforming standard PSO in simulations. Likewise, in experiments detailed in G. Zhao et al. (2022), an IPSO has been designed alongside traditional PSO and QPSO, focusing on position and accuracy, operation time, and algorithm convergence. The findings reveal that the approach introduced in this study has attained better accuracy in both orientation and position when compared to the other methods. Two PSO variants (standard PSO and IPSO) have been employed to compute the IK of a 7-DoF robot arm by Köker and Dereli (2018). The outcomes obtained with Global Local Best Inertia Weight and Random Inertia Weight are contrasted with those of the standard PSO. Results derived from computer simulations indicate that IPSO is proficiently applied for IKP.

Collinsm and Shen (2017) introduced a generalized fitness function for solving the full position and orientation IKP in

robotic manipulators with high DoF. It evaluates the effectiveness of PSO on simulated manipulators with multi-DoF and validates the approach on real-world hardware for a precision manipulation task. Luo et al. (2022) proposed a multi-objective full-parameter optimization particle swarm optimization (MOFOPSO) method to improve accuracy, efficiency, and stability in solving the IK of manipulators. The MOFOPSO considers position, posture, and joint factors while designing the multi-objective function, ensuring stability, high accuracy, and efficiency for a 6-DoF manipulator compared to six other IKS. In Adly and Abd-El-Hafiz (2016), parameters for single and multi-objective PSO have been proposed to solve the IKP. The paper provides a mathematical formulation of the problem and showcases applications on various robotic manipulators. The approach integrates a modified particle swarm optimization (MPSO) based on multi-objective optimization, where particles with higher fitness levels are updated towards the global best, while others adhere to a conventional movement pattern. Applied to solve the IK of a 5-DoF robotic arm, the suggested method shows superior performance compared to existing IKS. Umar et al. (2019) developed an innovative technique known as mutating particle swarm optimization (MuPSO) to address the

IKP in robotic manipulators. A forced mutation function was incorporated to prevent the algorithm from stagnating at local optima. Upon comparison with the existing approach, the MuPSO method demonstrated complete dominance in IKS over the alternatives. In Reyes and Gardini (2019), a PSO-based IKS has been suggested to solve the IKP of manipulators. This technique dynamically adjusts particle parameters, considers both PE and OE aspects, and it outperforms conventional IK algorithms. Khan, Abbasi, et al. (2020) developed the Dual PSO (DPSO) to solve the IKP of the manipulator. This solution employs the PSO algorithm in conjunction with the inverse Jacobian. In its distinctive application, the DPSO has partitioned the PSO algorithm into two components, optimizing trajectory orientation and position independently. The DPSO method accelerates convergence. Similarly, a DPSO technique for solving the IKP of robotic manipulators has been discussed by Khan, Kim, et al. (2020). Implemented in MATLAB, proves the effectiveness of the DPSO in resolving the IKP of a 6-DoF articulated robot arm with joint space constraints.

In Rokbani, Mirjalili, et al. (2022), a study was conducted to explore how PSO parameters have influenced the superiority and convergence of IKS. The results indicate that the PSO is a promising solver when employing suitable parameters and also the algorithm reveals statistical evidence of high-resolution convergence in terms of errors. In Karapetyan and Miryanova (2023), the PSO-based IKS was suggested for a multi-link robot. The recommended method was grounded in strategies for tackling the direct kinematics problem, sidestepping intricate computations, and avoiding singularity challenges. In F. Jin and Zhai (2020), sine-cosine particle swarm optimization (SCAPSO), a variant of PSO, efficiently solves the IK for a 6-DoF manipulator. It incorporates Sine and Cosine functions, preventing premature convergence and ensuring robust global search ability with fast convergence. Falconi et al. (2014) developed a Behavioral-based PSO (BPSO)-based IKS for addressing the IKP of serial manipulators. This BPSO algorithm categorizes the particle population into subgroups, each assigned a specific task, facilitating quicker convergence to the final result. In Rokbani et al. (2021), an innovative Beta PSO (β -PSO) method, leveraging Beta profiles for swarm exploration and exploitation, displayed exceptional performance in solving the IKP for a 6-axis generic robot arm. The β -PSO achieved an effective IK performance when compared to its counterparts. In de Lima Silveira Junior et al. (2018), a novel IKS named Fully Resampled PSO (FRPSO) has been presented to improve the PSO technique's performance in solving the IKP for robotic manipulators. The FRPSO reduces the problem dimension by half through complete particle resampling, eliminating the velocity vector and enhancing robustness against local minima. Comparative tests with three PSO versions on 4-DoF and 7-DoF robots establish the greater performance of FRPSO. Danaci et al. (2023) utilized PSO to resolve the IKP in serial robotic manipulators. Illustrated using the BAXTER robot with seven-joint arms, this algorithm is adaptable to any general robotic manipulator. In Gu et al. (2021), the inertia particle swarm optimization (InPSO) approach has been utilized to address the IKP for arm manipulators with 6-DoF. This method involves chaos initialization, inertia weight improvement, and particle variation. Comparative analysis with standard PSOs reveals that InPSO enhances the accuracy of the manipulator's IKS. In

Nguyen et al. (2022), the utilization of the PSO for solving the IKP in serial robotic manipulators has been explored. The PSO technique has verified the achievement of convergence for a complete end-effector pose and has been showcased through the application on the BAXTER robot. In Rokbani, Neji, et al. (2022), a novel algorithm based on PSO has been proposed. Alkayyali and Tutunji (2019) used PSO to efficiently solve the IKP for a 6-DoF KUKA KR 6 robot by minimizing end-effector pose errors by exploring the joint space. Simulations across three cases reveal the efficiency of the PSO-based IKS.

4.2.1.2 | Genetic Algorithm. The GA has been developed inspired by the concept of survival of the fittest (Darwin's theory of evolution) (Narayanan et al. 2022). Figure 12 illustrates the workflow of the GA. It begins with the random generation of an initial population, where each individual's joint configurations are encoded into chains. These encoded chains are decoded to represent the joint angles and configurations of the manipulator. Each individual's fitness is evaluated based on how well it achieves the desired end-effector position. The individual with the highest fitness is consistently selected and retained for the next generation, marking the selection process. In the next phase, crossover operations combine parent configurations to produce new offspring, while mutation operations introduce variations, generating a new generation. The process continues until optimal joint angle is arrived.

Sui et al. (2015) introduced an IK approach with high accuracy via Multiple Population Genetic Algorithm (MPGA). Multiple populations are employed to enhance convergence speed and mitigate the shortcomings associated with the least part point. The IK results of MPGA are compared to the Single-population Genetic Algorithm (SGA). Experiments conducted on the PANASONIC TA1400 robot validate the effectiveness of the proposed method. In Wu et al. (2020), an enhanced GA, referred to as Interval Search Genetic Algorithm Based on Trajectory (ISGABT), has been able to effectively address the IKP. The ISGABT incorporates three key strategies: population continuity, interval search, and reference point strategy. The technique was subsequently employed in solving the IKP for an 8-DoF robot. Experiments ensure motion continuity and accuracy. In L. Zhang, He, et al. (2017), the IKS utilizing Improved GA (IGA) has been presented. This approach primarily optimizes the joint space for a redundant robot. Additional focus was given to studying improvement strategies for IGA. The proposed IGA is more efficient than the conventional GA. Momani et al. (2016) investigated two GAs (conventional GA and continuous GA) to solve the IKP of robotic manipulators. The IKP is formulated as an optimization task to minimize deviation in the workspace. Simulation results demonstrate the continuous GA's superior IK performance when compared to the conventional GA. In Jaladi et al. (2020), the IK of serial manipulators has been addressed using a GA solver. The joint angles obtained from optimization techniques, including GA, algebraic, and PSO algorithms, are executed in ABB IRB 1600 serial manipulator with 6-DoF. The results indicate the superior performance of GA compared to its counterparts. In Krakhmalev and Korostelyov (2020), an IKS based on GA has been recommended. The optimization index (fitness function) is the square of the deviation of the position of the tool center point of the robot from the given trajectory point. The approach has

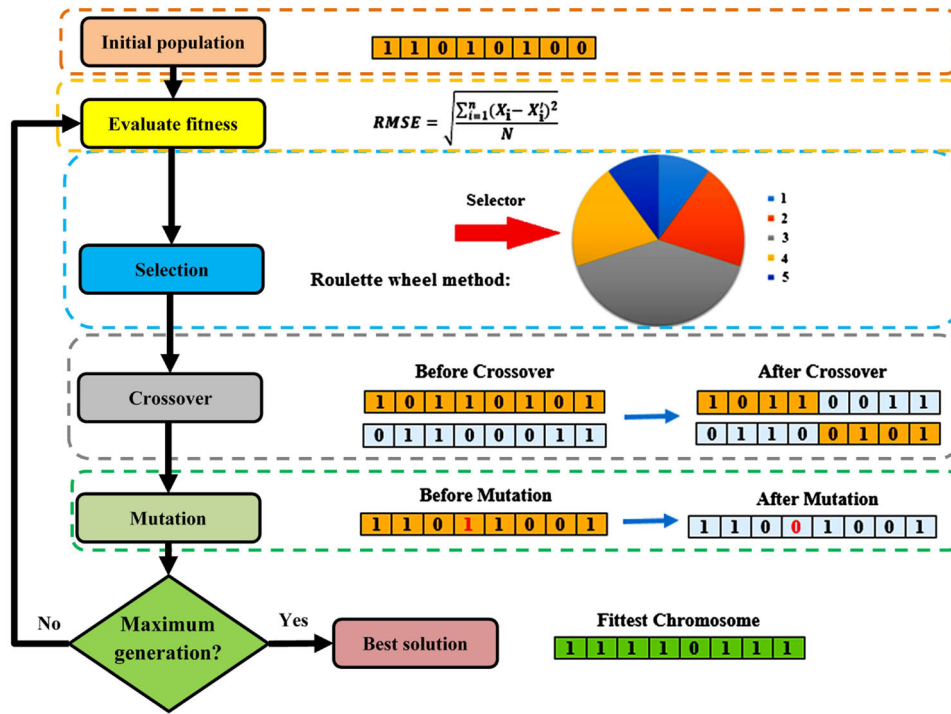


FIGURE 12 | Flowchart of the genetic algorithm (Mohamad et al. 2017). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70014)]

confirmed robust convergence and good accuracy. Dong et al. (2019) developed a GA-based IKS. This technique significantly diminishes the occurrence of robotic arm's angular momentum and provides better solutions. By employing a smooth reward function and a constraint function as optimization criteria, the IKP was effectively addressed.

4.2.1.3 | Other Metaheuristic Methods. Other metaheuristic techniques (FA, DE, GWO, ABC, etc.) that are used to solve the IK of the serial manipulators are provided below.

In Abdor-Sierra et al. (2022), a comparison of metaheuristic techniques for IKP in 6-DoF UR5 and 7-DoF MOTOMAN SIA20D robots has been presented. The study has validated each metaheuristic's robustness through 30 trials, assessing optimal solutions, standard deviation, and execution time to identify strengths and weaknesses. In Shinde and Pawar (2021), diverse metaheuristic algorithms have been applied to address the IKP, with their performance evaluated based on the end-effector's PE. The results indicate that Simulated Annealing (SA) exhibits superior performance compared to Sequential Quadratic Programming (SQP) and GA, showcasing minimal error in the end-effector position. J. Shi, Mao, et al. (2020) developed a Hybrid Mutation Fruit Fly Optimization Algorithm (HMFOA) for IKP. The HMFOA has combined olfactory and visual search strategies, addressing the premature convergence problem of the Fruit Fly Optimization Algorithm (FOA). Tested on a 7-DoF manipulator, HMFOA has outperformed other approaches in solving IKP. Soleimani and Ramli (Soleimani Amiri and Ramli 2021) introduced a novel method termed as hybrid Genetic-Swarm Optimization (GSO), which combines GA and PSO solutions to address the IKP in a robotic arm. The GSO technique has successfully solved the IK for each joint, and its performance has been analyzed through statistical analysis. The

results indicate that the GSO approach outperforms conventional methods. Similarly, in El-Sherbiny et al. (2018b), a knowledge-based artificial bee colony (K-ABC) solution has been developed to solve the IKP. Tested across various benchmarks, the K-ABC technique has exhibited exceptional IK performance (in a 5-DoF robot). Experimental results have revealed a significant 60% reduction in Mean Square Error (MSE) when compared to the conventional ABC. Lopez-Franco et al. (2018) have evaluated several metaheuristic approaches (ABC, Bat Algorithm (BA), Covariance Matrix Adaptation Evolution Strategy (CMA-ES), Cuckoo Search (CS), DE, differential search, GA, and PSO) for solving manipulator IKP. Among them, CMA-ES has outperformed, excelling in PE, OE, and execution time. In Rokbani, Mirjalili, et al. (2022), an IKS using Beta-distributed Salp Swarm Algorithm (β -SSA) was established and compared to the Salp Swarm Algorithm (SSA), QPSO, Bidirectional PSO (BiPSO), conventional ABC, and FA. Experimental results of IK solvers in 8-DoF link and KUKA KR 5 are tabulated in Table 2. In both manipulators, the β -SSA has shown an effective IKS.

Pena et al. (2016) utilized Non-Dominated Sorting Genetic Algorithm (NSGA II) and bacterial chemotaxis multi-objective algorithm to solve the IKP for a 6-DoF serial robot. Using the MOTOMAN MH6 robot model and a CAD-based workpiece, the study has minimized PE and OE through defined objective functions. In Mahanta et al. (2019), several metaheuristic techniques, namely ABC, PSO, Invasive weed optimization, and FA, were utilized to solve the IKP of 6-DoF KAWASAKI RS06L. Thorough simulation preceded experimentation, the IKS developed by the metaheuristic methods were within acceptable limits. In Slim et al. (2021), the Rough Sets Crow Search Algorithm (RCSA) solved IKP for the KUKA-KR16 manipulator. The RCSA has outperformed standard PSO, CS, FA, and

TABLE 2 | Performance of IKS.

Manipulator	Method	θ_1 (deg)	θ_2 (deg)	θ_3 (deg)	θ_4 (deg)	θ_5 (deg)	θ_6 (deg)	θ_7 (deg)	θ_8 (deg)	Error (m)
8-DoF link	β -SSA	-22.09	84.24	-36.88	-48.15	64.51	42.09	-35.63	43.77	7.343×10^{-25}
	SSA	-22.00	27.45	35.82	11.69	-40.32	45.10	60.35	-9.27	1.928×10^{-8}
	BiPSO (Ram et al. 2019)	-21.98	47.24	42.93	12.58	-16.83	-74.18	21.07	-36.93	1.791×10^{-8}
	FA (Rokbani et al. 2015)	-22.11	35.64	35.34	48.71	-29.67	-33.36	-40.88	-4.10	2.786×10^{-15}
	FA (Dereli and Köker 2020a)	-22.05	100.99	-62.96	45.76	9.52	-75.03	-0.11	2.22	1.835×10^{-15}
	QPSO (Dereli and Köker 2020a)	-22.07	40.71	-7.34	61.15	-34.72	12.53	89.96	-88.92	1.097×10^{-11}
	ABC (El-Sherbiny et al. 2018b)	-22.10	62.05	7.38	-47.76	44.01	-21.05	114.13	-38.06	6.973×10^{-8}
	β -SSA	-98.95	67.17	102.92	-48.10	-111.79	158.39	—	—	3.085×10^{-24}
KUKA KR 5	SSA	-110.93	55.80	109.71	-13.39	125.64	-6.97	—	—	5.855×10^{-9}
	BiPSO (Ram et al. 2019)	-118.15	90.78	56.84	-121.17	123.13	-10.01	—	—	2.757×10^{-12}
	FA (Rokbani et al. 2015)	-101.33	97.54	37.90	-126.09	-140.17	332.10	—	—	3.521×10^{-16}
	FA (Dereli and Köker 2020a)	-104.73	99.53	37.12	66.56	162.99	174.60	—	—	1.906×10^{-6}
	QPSO (Dereli and Köker 2020a)	-100.98	95.09	48.72	123.92	139.11	-151.00	—	—	2.775×10^{-17}
	ABC (El-Sherbiny et al. 2018b)	-103.74	99.13	36.13	48.88	153.08	141.46	—	—	2.688×10^{-15}

QPSO in providing more accurate IKS. Kanagaraj et al. (2022) compared the BA, Gravitational Search Algorithm (GSA), PSO, and Whale Optimization Algorithm (WOA) to solve the IKP. Simulation results have shown that BA and GSA have outperformed PSO and WOA. In Jangid and Kumar (2022), the Ant Lion Optimizer (ALO) has excelled in solving the IK of a 3-link manipulator, outperforming PSO, GWO, and Sine-Cosine Algorithm (SCA). The ALO's precision has been evident with a PE of 6.557×10^{-6} m and a solution time of 0.88 s, surpassing PSO, GWO, and SCA. Vaishnavi et al. (2021) introduced IKS for a 5-DoF manipulator, employing 14 heuristic techniques. The DE algorithm has outperformed with PE (4.425×10^{-8} cm) and OE (4.426×10^{-8} rad), while the GWO has excelled in computation time (0.308 s). Tsai et al. (2014) proposed a novel IKS for a 7-DoF manipulator. The proposed IKS, a combination of PSO and real-coded GA, has efficiently computed joint angles for seamless movement between points. This study has provided a comprehensive approach for IKP in manipulators. In M. Li et al. (2023), a hybrid IK method based on DE and CS known as Bidirectional Opposite Differential Evolution-Cuckoo Search (BODE-CS) has been suggested for the 6-DoF robot. Several schemes have been deployed to improve performance (accelerate convergence and avoid local optima). Experimental results have revealed that the technique is accurate and has fast convergence. In Seyyedabbasi (2022), a Sand Cat Swarm Optimization (SCSO) approach has been presented for the IKS of a PUMA robot arm. The SCSO is conceptualized by the swarms of sand cats. Among their primary actions, sand cats hunt for food and attack their prey. The SCSO is based on the ability of sand cats to detect low-frequency noises. The performance of the suggested method in solving the IK has been better than its other techniques. Yin et al. (2022) proposed an Equilibrium Optimizer Slime Mould Algorithm (EOSMA) and Multi-Objective EOSMA (MOEOSMA) for the IKS of a 7-DoF manipulator. Two scenarios have been considered, and the proposed algorithm has been compared with 15 single-objective and 9 multi-objective algorithms. The results have shown that the MOEOSMA has higher accuracy and shorter computation time. In H.-C. Huang et al. (2016), an Artificial Immune System tuned with Taguchi-Genetic Algorithm (AISTGA) has been suggested for the IKS of a 5-DoF manipulator. The AISTGA solver has outperformed conventional IK solvers, such as geometric, Jacobian, and PSO.

Further, a list of other metaheuristic methods established to solve the IK of serial manipulator is given in Table 3.

4.2.2 | Machine Learning Methods

4.2.2.1 | Artificial Neural Network. The ANN are complex structures, drawing inspiration from biological neurons, and offer a valuable method for tackling problems that cannot be effectively described analytically (Narayanan and Mathew 2022). A detailed overview of ANN (with learning methods, crucial components, basic architecture (with input, hidden, and output layers) and few types) is provided in Figure 13a. The ANNs are composed of fundamental processing units known as neurons (as shown in Figure 13b), which are connected to each other through weighted connections. By tuning the weights of the connecting neurons and utilizing appropriate activation functions, the relationship between

the input and output can be characterized. The way to implement supervised ANN is provided in the Figure 13c. Most of the ANNs learn the IK through training data, providing reliable solutions without mathematical modeling.

Şegota et al. (2021) developed a Multi-Layer Perceptron Neural Network (MLPNN) to solve the IK of a manipulator. Initially, the FK using the DH technique is determined, and a data set of 15,000 points has been generated. Subsequently, multiple MLPNNs have been trained with 10,240 different hyperparameter combinations to identify the best configuration. The MLPNN exhibits effective IK performance. In Gao (2020), an IKS based on the Improved Back Propagation Neural Network (IBPNN) incorporating an excitation function for a 6-DoF manipulator has been suggested. Results are compared to the conventional Back Propagation Neural Network (BPNN), revealing effective convergence and minimal error for the proposed IBPNN over the conventional BPNN. Ibarra-Perez et al. (2021) applied a Taguchi-based robust design to optimize BPNN parameters for the 6-DoF robotic manipulator IKS. The optimized parameters, encompassing hidden layers, neurons, learning rate, momentum, and data set size ratio, lead to a 90% prediction accuracy with less than a 5% margin of error during testing. This study (Borboni and Shakhovska 2023) introduces BPNN to solve the IKP. The BPNN was tested on a redundant manipulator, yielding results with acceptable accuracy and precision compatible with standard industrial applications. Wagaa et al. (2023) developed various ML algorithms including ANN, Bidirectional Long-Short Term Memory (BiLSTM), Long-Short Term Memory (LSTM), Convolutional Neural Network (CNN), Gated Recurrent Unit (GRU), and for solving the IKP of a 6-DoF manipulator. The outcomes are linked to the precise adjustment of Deep Learning Neural Networks (DLNNs) (BiLSTM, LSTM, GRU, CNN) hyperparameters applied across diverse DoF manipulators as given in Table 4.

Köker et al. (2014) have suggested a Neural Network Committee Machine (NNCM)-based IKS for a 6-DoF redundant robotic manipulator. Ten ANNs were designed to develop the NNCM, which has been trained with specific data sets selected based on direct kinematics, thereby enhancing the overall IK performance. In Ma et al. (2016), a Radial Basis Function Neural Network (RBFNN) has been implemented to address the IKP of a 6 R serial manipulator. The final simulation results reveal that accurate IKS can be obtained more efficiently using RBFNN. Additionally, practical implementation demonstrates its effective IK performance. Handayani et al. (2018) employed the Bayesian regularization backpropagation (BP) training function to train ANNs for solving the IK of a planar manipulator. The robot executes nodes resembling square and triangle shapes in its workspace based on the ANN solution. The outcomes highlight the effective performance of ANNs in resolving the IK of the planar manipulator. A modified Generative Adversarial Neural Network (GANN) for real-world IK using experimental data has been developed in H. Ren and Ben-Tzvi (2020). By overcoming the constraint of extensive training data sets, the technique introduced in this study utilizes generator and discriminator architectures, surpassing state-of-the-art algorithms in scenarios with limited data. The assessment conducted on MICO and FETCH robotic manipulators using diverse training data sets validates the efficacy of GANN in delivering accurate

TABLE 3 | Other metaheuristic methods deployed for IKP of serial manipulators.

Reference	Year	Metaheuristic methods	Manipulator	DoF	Validation Simulation/ Experiment	Inference
Sheik Masthan et al. (2023)	2023	Improved GSA	YASKAWA MH5	6	Experiment	Noted maximum PE was 1.661×10^{-8} m.
Slim et al. (2023)	2023	BA	DOBOT	4	Experiment	The observed PE was in between 1.14×10^{-14} cm and 9.61×10^{-14} cm.
Nyong-Bassey and Martyns Epemu (2022)	2022	GWO, improved GWO, PSO, jellyfish, WOA	—	6	Simulation	The simulation results of the algorithms considering PE as a performance index demonstrated that GWO performs superior over other methods.
Vazquez-Castillo et al. (2022)	2022	DE	ABB IRB-1600	6	Experiment	Satisfactory IK results were noted.
Wu et al. (2021)	2021	Multi-objective GA	—	8	Simulation	The recommended IKS performed better than its counterparts.
Tam et al. (2021)	2021	DE	—	5	Simulation	In the analysis of three case studies, it has been demonstrated that DE efficiently tackled the IKP, resulting in minimal errors.
Nguyen (2021)	2021	Improved DE	—	7	Experiment	IKS arrived via improved DE was better than IPSO.
Hernandez-Barragan et al. (2021)	2021	Improved FA	—	5	Simulation	The suggested approach outperformed FA in IKS.
Dereli and Köker (2020a)	2020	FA	SUNGUR 370	7	Simulation	Several simulation studies were conducted and the PE ranged between 5.59×10^{-11} and 8.77×10^{-3} cm.
Dereli and Köker (2020b)	2020	ABC	SUNGUR 370	7	Simulation	Compared with PSO, the ABC technique's effectiveness in PE and computation time was confirmed.
Sahu et al. (2020)	2020	Modified FA	—	4	Simulation	Acquired IKS are compared to CS, FA, ABC, and BA. The suggested algorithm surpassed its counterparts.
Urrea and Saa (2020)	2020	SA	—	6	Simulation	Significant IK performance was observed.
(Nguyen et al. 2020)	2020	DE, PSO, IPSO, Improved DE	—	5	Simulation	The IPSO and improved DE showed effective IKS (better accuracy and less computation time) than other methods.
L. Zhang and Xiao (2019)	2019	Improved ABC	CYTON GAMMA 300	7	Simulation	On comparing with GA, PSO, and ABC, the suggested method shows effective IK results.

(Continues)

TABLE 3 | (Continued)

Reference	Year	Metaheuristic methods	Manipulator	DoF	Validation Simulation/Experiment	Inference
Antonio-Gopar et al. (2018)	2018	DE	BAXTER	7	Simulation	The calculated PE and OE are 0.04 mm and 0.2 deg, respectively.
Mao et al. (2017)	2017	DE + MPMO	—	6 & 10	Simulation	Compared to DE, PMO, MPMO, ACO, and ABC, the recommended IKS showed effective results.
Costa et al. (2016)	2016	Improved SA + NSGA II	—	12	Simulation	The IKS was successfully acquired and the results were satisfactory.
Rokbani et al. (2015)	2015	FA	—	3	Simulation	The FA showed better convergence time (average 2.339×10^{-3} s) and PE (3.116×10^{-8} m).

IKS. Sari (2014) has examined the impact of the learning approach on an ANN for IK of a seven-joint redundant robotic manipulator. Further investigation into ANN types, activation functions, and hidden layer neurons using the Levenberg-Marquardt (LM) algorithm reveals optimal IK performance was achieved with an FFNN, logistic sigmoid activation (logsig), and 41 hidden layer neurons.

Kumar and Chand (2015) presented the IKS for the SCORBOT-ER 4u robot arm (5-DoF) by using a multi-Layered Feed-Forward Neural Network (MLFFNN). Data generation and MLFFNN training have been accomplished by utilizing DH and geometrical techniques. The algorithm has undergone testing on SCORBOT-ER 4u and confirmed the efficacy of the suggested approach. In Srisuk et al. (2017), ANN-based IKS has been employed for a robotic arm. The 4-layer ANN represents end-effector positions and the BP with a variable learning rate has refined weights until convergence, resulting in optimal weights for the IKS. Simulations in MATLAB have exhibited the method's efficiency, enabling the robotic arm to accurately reach the desired arm angle. Raj et al. (2015) has constructed a Feed-Forward Backpropagation Neural Network (FFBPNN) and RBFNN-based IKS for a 5-DoF manipulator. Performance comparison has revealed that RBFNN has outperformed FFBPNN, yielding minimal error. In Soylak et al. (2017), the IKS for a plasma cutting manipulator (3-DoF) has been provided by an ANN (with LM technique) trained with joint angle values. Validation with a new test data set and simulation has confirmed the accuracy of the plasma cutting torch. Dalmedico et al. (2018) have developed an ANN for 4-DoF IK in 3D with two configurations: the first with 40 neurons (10 per layer) and the second with 60 neurons (15 per layer). Both configurations have achieved the desired precision through training. However, the Euclidean errors averaged to be 0.112 and 0.219 cm for the first and second configurations, respectively. Shaar and Ghaeb (2023) suggested a Recurrent Neural Network (RNN)-based IKS. This method was applied to a 6-DoF manipulator. The recommended approach has been compared to other ANN techniques. The simulation study reveals that the performance of the RNN was better than other ANNs.

Further, other ANN methods deployed in solving the IKP of the serial manipulators are tabulated in Table 5.

4.2.2.2 | Adaptive Neuro-Fuzzy Inference System. The architecture of the ANFIS (as displayed in Figure 14) consists of five layers known as Layer 1—Fuzzification layer, Layer 2—Participation layer or the product layer, Layer 3—Normalization layer, Layer 4—Defuzzification layer, and Layer 5—Output layer. The ANFIS integrates human experience and knowledge through fuzzy logic and leverages the learning capability of ANN (Narayanan and Ramakrishnan 2021). By training Membership Functions (MFs) using ANN, fuzzy rules are derived from the data set and MFs are trained accordingly. As a result, the trained rules can predict the output for a given input.

Vladimirov and Koceski (2019) addressed IKS using an ANFIS for a 2-DoF robot arm. A comparative analysis with an analytical method reveals the effectiveness of the ANFIS in achieving better IKS. The utilization of ANFIS to solve IKP has been showcased in Hussein et al. (2020), where a 4-DoF robot

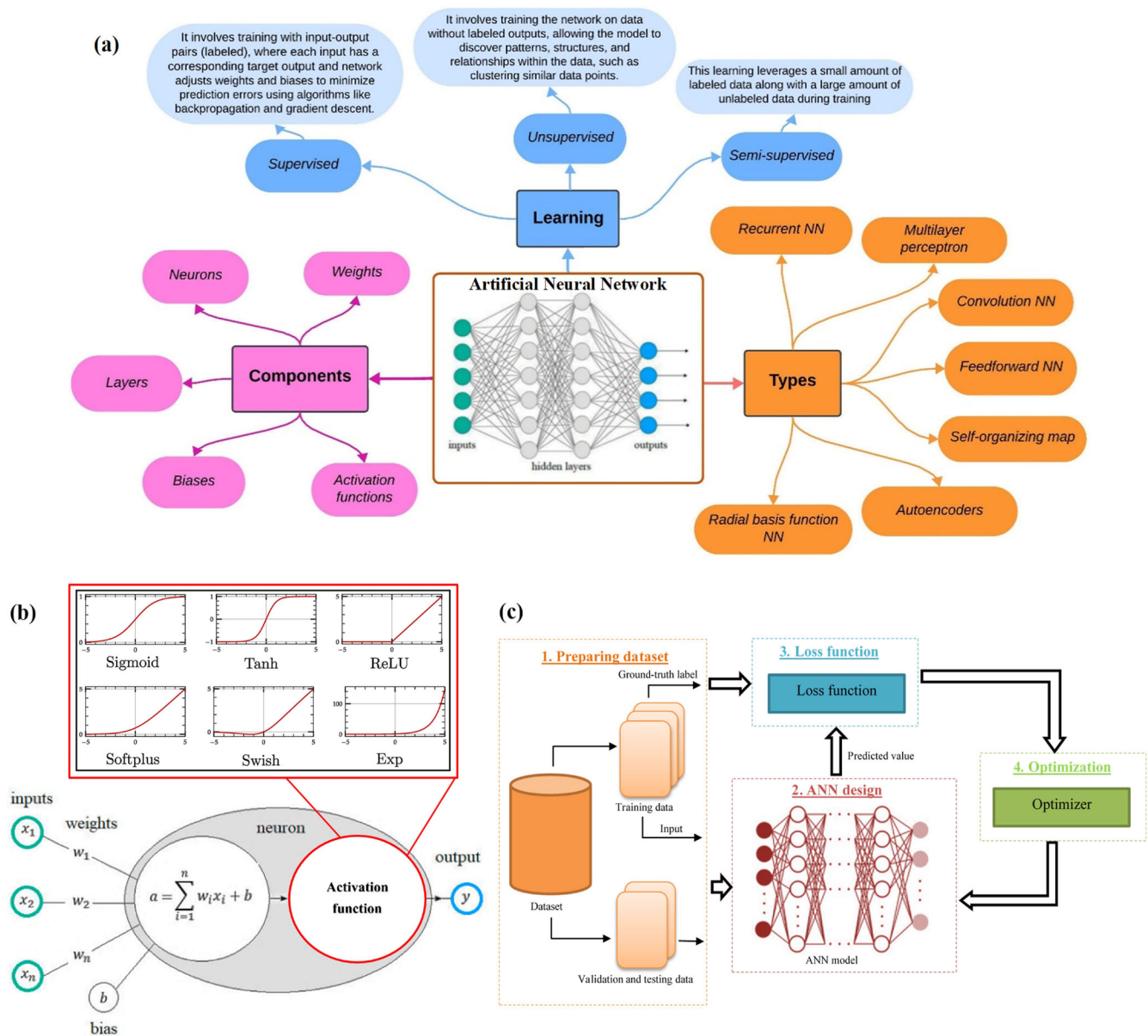


FIGURE 13 | Artificial neural network: (a) Overview of ANN, (b) neuron (Dropka and Holena 2020; Pantalé 2023), and (c) implementation of supervised ANN. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

manipulator (IRIS robot) serves as the IK model. Initially, the FK has been derived, and then ANFIS technique provides IKS by learning from training data. Simulation results demonstrate the ANFIS's capability to produce acceptable IKS. Tho et al. (2015) proposed an ANFIS methodology to solve the IKP. The method employs the least square estimation technique for tuning linear output MF parameters and BP algorithm for tuning nonlinear input MF parameters, which results in the effective IKS. The use of ANFIS to solve IKP has been presented in Hendarto et al. (2014). The PUMA robot arm virtual model was utilized, and by generating an ANFIS approach (in MATLAB) with three different MFs, the impact of MF number on PE was explored. Increasing the number of MFs results in a reduction in PE and the paper reports a PE of 27.974 mm while using 10 MFs. The study in Deshmukh et al. (2021) has aimed to simplify the IK by employing ANFIS. The manipulator has been mathematically modeled in MATLAB, and its IK analysis

has been executed. ANFIS has been trained using data generated from the mathematical model. The results are validated through simulation studies. In Das and Mahapatra (2015), the utilization of ANFIS is evidenced in solving the IKP through generated data. A single-output Sugeno-type Fuzzy Inference System (FIS) with grid partitioning has been constructed for this purpose. The analysis of IK for a 5-DoF manipulator has been thoroughly examined. Similarly, Das et al. (2014) utilized the ability of ANFIS to predict the IKS of the manipulator, and single-output Sugeno-type FIS employing grid partitioning was modeled. The IK for 2-DoF, 3-DoF, and 5-DoF robot manipulators are symmetrically analyzed to show the effectiveness of this technique. Omer et al. (2018) utilized the ANFIS to achieve the solution for the IK of the PUMA robot. The use of ANFIS has proven highly efficient in addressing accuracy and precision issues. The results obtained in this study establish a decrease in error by 0.21% when compared to its counterpart. Abbas et al. (2019)

TABLE 4 | Various IKS of ANNs (Wagaa et al. 2023).

DoF	Neural Network	Time (s)	Performance (%)	PE (mm)	OE (deg)
3	BiLSTM	0.334	99.96	0.0004	0.432
	GRU	0.331	99.87	0.0016	0.856
	LSTM	0.329	99.92	0.0006	0.542
	CNN	0.327	99.91	0.0008	0.586
	ANN	0.328	99.91	0.0009	0.640
5	BiLSTM	0.583	99.08	0.0010	0.765
	GRU	0.331	97.85	0.0016	0.856
	LSTM	0.584	97.88	0.0020	0.813
	CNN	0.574	97.87	0.0030	0.992
	ANN	0.538	97.85	0.0030	0.998
6	BiLSTM	0.659	95.35	0.0040	0.847
	GRU	0.642	94.05	0.0070	1.121
	LSTM	0.654	94.91	0.0030	0.989
	CNN	0.637	94.64	0.0040	0.994
	ANN	0.633	94.48	0.0050	0.998

implemented the ANFIS technique to compute the IKS. The recommended approach has been deployed in a 5-DoF manipulator to test its performance. Simulation analysis has showcased promising outcomes. Furthermore, in Marzouk et al. (2018), ANFIS has been utilized to determine the IKS for a 5-DoF manipulator. The ANFIS model accurately specifies the coordinates aligned with specific angles of cartesian coordinates. The average error of the training data set for the joint coordinates θ_1 , θ_2 , θ_3 , θ_4 , and θ_5 are recorded as 2.801×10^{-5} , 4.19×10^{-6} , 3.31×10^{-5} , 3.72×10^{-5} , and 7.33×10^{-6} , respectively.

In Refaai (2021), it has been concluded that using the ANFIS method for solving the IKP robotic manipulator results in a significantly lower computational cost. A noteworthy aspect of employing the ANFIS technique is the avoidance of complex equations. Utilizing ANFIS demonstrates a commendable and acceptable training error. Kumar et al. (2022) employed ANFIS to resolve the IK of the KUKA manipulator. Simulation studies were conducted using MATLAB, and the 3D model of the KUKA manipulator was depicted using an ROBO analyzer. The simulation results were computed, showcasing the error difference for various joint angles. Jiang-nan et al. (2015) introduced an ANFIS-based IKS for a 4-DoF manipulator. Joint space curves are obtained using cubic splines, and ANFIS is trained with randomly selected sample points from the curve. Results indicate that ANFIS had better IKS in terms of better accuracy and lower computing time, outperforming the BPNN. In Singh and Banga (2020), the IKS for the PUMA robot was determined using ANFIS. The difference between the analytical and predicted values was calculated, and the results are under acceptable limits and quicker than other methods. Narayan et al. (2018) demonstrated IKS (5-DoF robot) using ANFIS. The ANFIS predicts joint parameters corresponding to the arm matrix for the home position of the manipulator. The results show that the estimated error values are relatively lesser and satisfactory. Narayan and Singla (2017) presented ANFIS for

solving the IKP of a 4-DoF manipulator. Here, the ANFIS was designed with a Gaussian MF. The analytical method was developed as a comparison in the MATLAB environment, and simulation results reveal the satisfactory outcomes of the ANFIS. In Manjaree et al. (2016), the ANFIS method, employing two different MFs, has been applied for IK analysis. Experiments are conducted to validate the performance of the suggested IKS. Further, the performance of the ANFIS is compared to the analytical approach to verify its significance. Promising results are observed in ANFIS. The IK analysis of a 5-DoF industrial robotic using ANFIS has been presented in Manjaree et al. (2015). The ANFIS was studied with two different MFs, generalized bell and Gaussian, and their performances are compared to the analytical technique. An experiment is conducted, and the results underscore the significance of selecting an MF for manipulators. Asadi-Eydivand et al. (2015) suggested an ANFIS to resolve the IKP. Through the simulation study conducted in MATLAB 2008a, it has been revealed that the performance of the proposed algorithm meets satisfactory limits and is applicable for manipulators with higher DoF.

4.2.2.3 | Reinforcement Learning. The RL operates by rewarding favorable behaviors and/or penalizing unfavorable ones (Narayanan 2023). An RL agent typically possesses the ability to perceive and comprehend its environment, take actions, and learn from mistakes. The RL mechanism is illustrated in Figure 15, and Table 6 provides an overview of different RL methods.

Blinov et al. (2021) introduced the DQL algorithm for the IKP of a four-link manipulator. Employing a dynamic exploration coefficient and a novel Q-table generation approach, the prevents algorithm convergence to local optima, reducing training time and hardware requirements. Tested in three environments of varying complexity, the method demonstrates successful training across scenarios, highlighting its adaptability to diverse

TABLE 5 | Various other ANN-based IKS for serial manipulators.

Reference	Year	ANN Methods	Manipulator	DoF	Validation Simulation/ Experiment	Inference
Ho et al. (2023)	2023	DLNN	FRANKA EMIKA PANDA	7	Experiment	In comparison to other methods, the PE and OE improved by 36% and 48%, respectively.
Bahani et al. (2023)	2023	MLPNN	ABB IRB 120	6	Simulation	The IK results of MLPNN were better than geometric and algebraic techniques.
Vu et al. (2023)	2023	Learning-based framework	KUKA LBR IIWA	7	Experiment	The ANN directly learns optimal solutions, ensuring accuracy and real-time capability (32 μ s) on the KUKA manipulator.
Lu et al. (2022)	2022	MLPNN	XARM6	6	Simulation	Observed PE and OE are 0.001 mm and 0.001 rad, respectively.
Vaishnavi et al. (2022)	2022	DLNN	—	5	Simulation	The performance of DLNN was superior to LSTM and GRU in providing IKS.
Anh and Bien (2021)	2021	DLNN	—	6	Experiment	The performance of the suggested algorithm was satisfactory.
Peng et al. (2021)	2021	FFNN	ABB IRB7600-500	6	Simulation	The IK performance of the recommended approach was better than other methods.
Semwal and Gupta (2022)	2021	MLFFNN	—	3	Simulation	The suggested MLFFNN surpassed the model-driven IKS.
Aravindhakshan et al. (2021)	2021	FFNN	—	5	Simulation	The proposed algorithm improved the IK performance when compared to conventional methods.
Kamlesh Shah et al. (2020)	2020	DLNN	—	5	Experiment	Experimental results reveal that there was a deviation of ± 0.1 rad in each joint.
Elkholy et al. (2020)	2020	CNN	—	7	Simulation	The CNN was trained using MATLAB-generated data, incorporating DH parameters and manipulator workspace. The performance of CNN was healthier than its counterparts.
Shah and Bhasin (2021)	2020	Neural-Net	—	4	Experiment	Its efficient structure allows execution on the NVIDIA Jetson Nano module.
Bhardwaj et al. (2019)	2019	Unsupervised ANN	PUMA	6	Simulation	Improved IK performance was noted even after considering fewer data set.
Takatani et al. (2019)	2019	FFBPN	—	3	Simulation	The successful learning of the IK ANN model in this study has occurred without the utilization of the gradient or structure of the evaluation function.
Julian Carini Montenegro et al. (2018)	2018	MLFFNN	—	3	Simulation	The MLFFNN surpassed the performance of the inverse Jacobian technique.

(Continues)

TABLE 5 | (Continued)

Reference	Year	ANN Methods	Manipulator	DoF	Validation Simulation/ Experiment	Inference
Hargis et al. (2018)	2018	FFNN	AUBO-i5	6	Experiment	The results demonstrate that ANN is capable of achieving accuracy down to the millimeter level.
Mahajan et al. (2017)	2017	Unsupervised ANN	—	2	Simulation	Effective performance was noted even by training with a limited data set.
N. Zhang, Zhang, et al. (2017)	2017	BPNN	RBT-6T/S03S	6	Simulation	The IK performance of BPNN was under acceptable limits.
H. Li et al. (2017)	2017	BPNN	RS10N	6	Simulation	The PE noted was between -0.04 mm and 0.01 mm.
Csiszar et al. (2017)	2017	FFNN	—	3	Simulation	The IK errors were within the acceptable limits.
Nagata and Watanabe (2016)	2016	FFNN	MOTOMAN	6	Simulation	The proposed technique was able to learn from the data set effectively.
Putra (2016)	2016	ANN	—	3	Experiment	The IK models of the robot had an error under 2%.
Nagata et al. (2016)	2016	FFNN	MOTOMAN HP20	6	Simulation	Effective IK performance was observed for the proposed algorithm.
Almusawi et al. (2016)	2016	ANN	DENSO ROBOTIC ARM	6	Experiment	The developed approach outperforms existing IKS in the literature.
Aggarwal et al. (2014)	2014	MLFFNN	—	6	Simulation	It has been found that MLFFNN gives a minimum error and effective IK results.
Yuan and Feng (2014)	2014	RBFNN	PUMA	6	Simulation	The method effectively minimizes system errors, providing a solution to model-driven IKS drawbacks like computational load, slow convergence, and low accuracy.

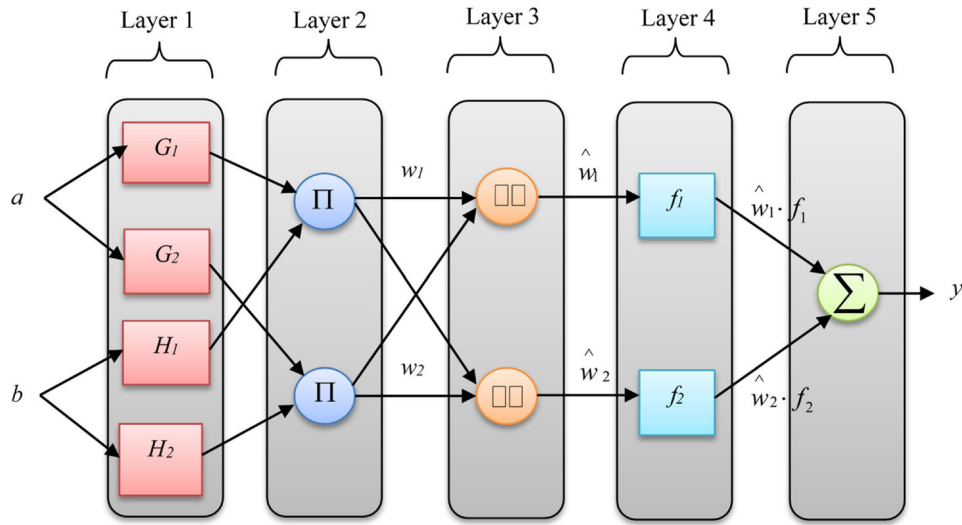


FIGURE 14 | ANFIS architecture (Narayanan and Ramakrishnan 2021). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.20014)]

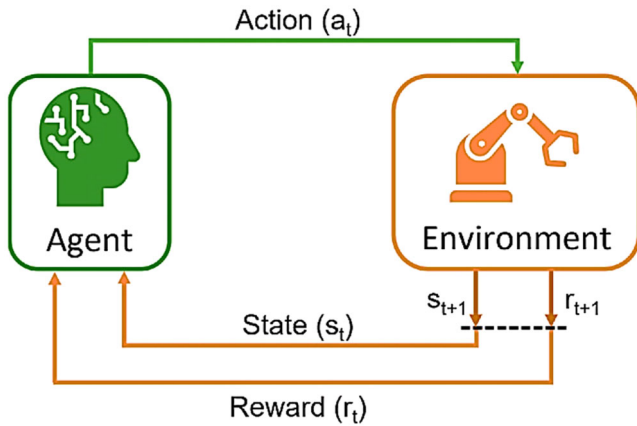


FIGURE 15 | Reinforcement learning framework (del Real Torres et al. 2022). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.20014)]

IKP and manipulator configurations. In addressing the IKP of a serial manipulator, Weber and Schmidt (2021) utilized a DDPG agent, surpassing other state-of-the-art techniques for arm robots. It achieves a remarkable 99.4% success rate with a mean error distance of 0.015 m. Malik et al. (2022) developed a DQN as an IKS for the 7-DoF manipulator. Experimental implementation on a SAWYER arm proves DQN's ability to learn diverse trajectories accurately, showcasing its adaptability in various scenarios. Guo et al. (2019) designed DDPG to acquire the IKS for a 5-DoF arm robot with a gripper. Simulations conducted using Tensorflow and Matplotlib aimed to validate the precision of this innovative approach. Simulation results reveal that, when compared to model-driven methods, the error falls within an acceptable range, affirming the satisfactory accuracy of both angle and posture. Schmitz and Berthet-Rayne (2021) explored the utilization of RL and PPO to address the IKP of a 5-DoF manipulator. The outcomes indicate that the suggested solution excels in learning the kinematic model of a 5-DoF system compared to the DLS technique, showcasing improvements in computation time, PE, and OE. Besides, a Deep Reinforcement Learning (DRL)-based algorithm has been deployed in Rahul and Chiddarwar (2023), incorporating IK with a time-varying gain module. This novel approach has

confirmed remarkable performance improvements in terms of robustness, optimality, and convergence speed, when compared to model-driven methods, achieving an outstanding average success rate of 97%. In X. Shi, Guo, et al. (2020b), an innovative solution employs an RL-TD network constructed with PyTorch to address IK for a 5-DoF arm robot. Enhanced distributed multiple rewards, based on the position difference between adjacent joints, optimize the technique, preventing unreachable points and ensuring mechanical structure safety. The algorithm's effectiveness is confirmed through simulation studies. Emami et al. (2022) employed the multiagent reinforcement learning (MARL) approach for solving the IKP in manipulators. Commencing with foundational concepts, the authors progress through advanced methods, underscoring the intrinsic advantages of MARL in tackling challenges within robotics research. The paper ultimately delves into the choice between model-based or model-free IK approaches, encompassing varied applications of value-function-based or policy-search techniques.

4.2.3 | Other Data-Driven Methods

Other data-driven techniques deployed to solve the IKP of serial manipulators are examined in detail in this section.

In P. Jha et al. (2015), a hybrid Multi-Layered Perceptron Gravitational Search Algorithm (MLPGSA) has been established to tackle the IKP of the PUMA manipulator. A comparison with the Multi-Layered Perceptron Back Propagation (MLPBP) reveals that MLPGSA delivers better IK results with minimal error. Hassan et al. (2020) directed their attention towards solving the IKP of manipulators, approaching it as an optimization problem within Quadratic Programming (QP) and addressing it with the utilization of RNNs. Various case studies have been analyzed, and the IK results demonstrated by the RNNs are more effective than other methods. In this study (Zayer et al. 2021), the IKS of a 2-DoF manipulator has been arrived using Fuzzy Neural Petri Net (FNPN). The MATLAB environment has been used to simulate the IK equations, and the necessary data set has been collected to train the FNPN. Effective IKS has been obtained while using FNPN. In Köker

TABLE 6 | Various reinforcement learning algorithms (Tong et al. 2023).

RL methods	Off-Policy/On-Policy	Learning task
Deterministic Policy Gradient (DPG)	Off-policy	Interpolating between policy optimization and optimal value function
Deep Deterministic Policy Gradient (DDPG)	Off-policy	Interpolating between policy optimization and optimal value function
Proximal Policy Optimization (PPO)	On-policy	Optimal policy
Trust Region Policy Optimization (TRPO)	On-policy	Optimal policy
Asynchronous Actor-Critic (A3C)	On-policy	Optimal policy
Double Q-Learning (DQL)	Off-policy	Optimal value function
Deep Q-Network (DQN)	Off-policy	Optimal value function
Q-learning	Off-policy	Optimal value function
Q (λ)	Off-policy	Optimal value function
State Action Reward State Action (SARSA)	On-policy	Optimal value function
Monte Carlo (MC)	On-policy	State value
Temporal difference (TD)	On-policy	State value
Value iteration	—	State value
Policy iteration	—	State value

and Çakar (2016), hybrid IKS using ANN, GAs, and SA has been studied. The hybrid IKS has achieved a micrometer-level reduction in end-effector error. Shastri et al. (2020) deployed an ANN for IKP in a robot manipulator. Here PSO, Segmented PSO (SPSO), and Modified SPSO (MSPSO) have been used to optimize ANN parameters. Comparisons based on fitness function and regression have revealed that MSPSO outperforms, with a fitness function value of 6.18×10^{-9} , compared to 2.35×10^{-3} (by PSO) and 1.10×10^{-5} (by SPSO). Bai et al. (2021) have suggested an optimized BPNN using the FOA for IKP. Overcoming issues of low convergence accuracy and speed, it has outperformed standard BPNN and PSO-optimized BPNN, resulting in a smaller error varying between -0.047 and 0.127 mm. Bayati (2015) deployed the CS algorithm and Imperialist competitive approach for solving IKP. On comparing the IK results with GA, PSO, harmony search, DE, ANFIS, MLPNN, and RBFNN, the suggested method has executed more effective IK performance than their counterparts. In Toshani and Farrokhi (2014), a novel online method incorporating ANN, fuzzy logic, and a QP has been developed to address the IK of the 7-DoF manipulator. This IKS employs an adaptive strategy based on the Lyapunov theory and Kuhn-Tucker conditions, yielding significant outcomes. Khaleel (2018) presented the solution for IKP for one of the 3-link robots by combining GA and ANN. The inputs are the position and orientation of the 3-link robot. Simulation results have shown that the performance of the recommended technique was within the acceptable limit. In the study (Al-Qurashi and Ziebart 2019), a DLNN has been unveiled for resolving the IKP of a complex robotic manipulator with high DoF. Moreover, the algorithm has been implemented to tackle the IKP of the BAXTER manipulator. Results obtained from the experiments demonstrate the ability of the BAXTER end-effector to achieve desired positions and orientations with precision, showcasing minimal error.

In J. Chen and Lau (2016), support vector regression (SVR) with ML has been blended to solve the IKP of redundant manipulators. The proposed IKS has been verified in 7-DoF MITSUBISHI PA-10 robot and results have proved its high accuracy and effectiveness. In Shihabudheen and Pillai (2015), an Evolutionary Fuzzy Extreme Learning Machine (EF-ELM) has been used to predict the IK of 2-DoF and 3-DoF robotic arms. A comparative analysis was conducted between the proposed and conventional Extreme Learning Machine (ELM) and BPNN. The proposed EF-ELM-based IK results surpassed ELM and BP methods. Mustafa et al. (2016) implemented SVR and Kohonen Self Organizing Map (KSOM) to solve the IKP of a 5-DoF manipulator. The SVR methods showed less computation time and KSOM had better accuracy. On the other hand, in Jiokou Kouabon et al. (2020), a novel method employing a Growing Neural Gas Network (GNGN) has been introduced for IKS. By conducting workspace and configuration space clustering of specific joints, the IKS incorporates a neighborhood function to enhance IK performance. Simulations and experimental trials conducted on manipulators have verified the efficacy of the GNGN approach. Olaru et al. (2014) suggested a novel method known as Cycle Coordinate Descent Method coupled to Sigmoid Bipolar Hyperbolic Tangent Neural Network (CCDM-SBHTNN) for IKP. Simulation results have revealed the phenomenal strength of the CCDM-SBHTNN in providing a relatively better IKS for a manipulator. In Y. Huang et al. (2023), a Back Propagation-Sequential Quadratic Programming (BP-SQP) for IK analysis of the UR10 robot has been developed. Here, ANN has been integrated into SQP to develop the BP-SQP. The findings indicate that while the suggested technique was quicker, it necessitates a specific number of samples as a criterion. The ML and GDLS-based IKS have been outlined in X. Wang et al. (2020c). This method has integrated an optimally enhanced step-size coefficient determined by

an ML model trained with extensive data points. The technique has efficiently converged in less than 10 iterations, underscoring its practicality and effectiveness. The application of DLNN combined with the DLS approach has been recommended for addressing the IKP in X. Wang et al. (2021). This integration has notably led to substantial improvements in algorithm convergence. Remarkably, the convergence rate reached an impressive 95.47% when the error threshold was set at 0.01 mm, surpassing the performance of numerous existing methods. Athulya et al. (2020) developed vision-based inverse kinematics (VIK)-based IKS for a 2-DoF robot. The camera has been placed at a known distance from the manipulator, and the coordinates have been measured from the photos taken by the camera. The MLPNN was used to find the IKS, and the results revealed the effective performance of the VIK. Similarly, Megalingam et al. (2022) executed an image processing-based IKS for a 5-DoF manipulator. A simulation has been conducted using MATLAB to generate arm configurations based on the coordinates acquired from the camera. The simulation results have confirmed effective IK outcomes. In W. Xu et al. (2017), three regression approaches such as K-Nearest Neighbors Regression (KNNR), Gaussian mixture regression, and ELM have been applied to provide IKS. Among these methods, the KNNR performed better with the lowest Root Mean Square Error (RMSE) of 2.127 mm.

Furthermore, Table 7 lists some other data-driven methods deployed for solving the IKP of the serial manipulators.

5 | Discussions: Challenges and Future Directions

This paper examines various IK approaches applicable to serial manipulators, offering extensive state-of-the-art insights. Several model-driven and data-driven methods are analyzed based on their key characteristics, advantages, disadvantages, and applicability. A detailed performance evaluation is given in Table 8, where each method is assessed across various parameters (memory requirement, convergence speed, interpretability, robustness, and scalability). To facilitate comparison, these parameters are categorized as low, medium, or high, providing a clear understanding of their effectiveness. A low rating indicates poor performance in a specific aspect, highlighting significant limitations or inefficiencies. Whereas, a medium rating represents moderate effectiveness or performance. Finally, a high rating signifies strong performance, demonstrating greater suitability for the concerned task. Overall, this comparison gives a road map for choosing the appropriate IK method. Subsequently, a critical analysis of each technique is provided in the following discussion.

Section 4.1 discusses model-driven methods, namely numerical, geometric, analytical, algebraic, and other model-driven methods. As seen in Figure 5, around 37% of the literature reviewed falls under this approach. Among these techniques, major publications were noted under the analytical method, and the least publications were observed in the algebraic method. This reveals that the very least preference is given to the algebraic-based IKS in the current research due to its complex methodology. Consequently, a significant reduction in publication is noted in the model-driven methods due to the overwhelming

improvement in technology which has inclined the research towards data-driven techniques (Starke et al. 2016). Moreover, in many cases, the performance of the model-driven IKS was not able to surpass the data-driven IK methods (Semwal and Gupta 2022; Guo et al. 2019; Rahul and Chiddarwar 2023). The hybrid model-driven methods (discussed in the other model-driven methods section) overcame the problems of the model-driven methods. This is accomplished by combining two model-driven methods, and each method's issues are compensated. Therefore, it makes the hybrid model-driven method to be more preferable for IKS. The work reported by M. Jin et al. (2020) has demonstrated that the hybrid model-driven method has surpassed the ANN-based IKS. Additionally, the FABRIK algorithm presented in Semwal et al. (2021) also surpassed the IK performance of ANN. Nevertheless, it is suggested that with improvements in the number of layers, neurons, and training data set, ANN may outperform the hybrid model-driven and FABRIK methods.

Consequently, Section 4.2 investigated various model-free/data-driven methods namely: metaheuristic (PSO, GA, and other metaheuristic methods), ML (ANN, ANFIS, and RL), and other data-driven methods. Nearly 63% of the literature utilized in this review covers data-driven methods which reveals that these methods are more preferred/suitable for solving the IKP of serial manipulators. The accuracy and adaptability of these methods are phenomenal when compared to other techniques (Singh et al. 2023). As pointed out by Semwal and Gupta (2022), the primary benefit of employing data-driven techniques for IKS is their capability to seamlessly accommodate any number of links, requiring minimal effort. In contrast, the other methods necessitate careful consideration of factors such as the number and types of links. Thus, it becomes more advantageous for data-driven, and so more publications are observed in data-driven techniques. Among data-driven methods, more publications are noted in ANN method. Even ANFIS falls under a data-driven approach since the MF are tuned by ANN methods. Subsequently, these methods deal effectively with nonlinearity through self-organization, self-learning capabilities, and a high degree of fault tolerance (Narayanan et al. 2022). However, the IK performance of SA was relatively lower than other data-driven methods.

Moreover, when multiple IKS exist for the same task space coordinates, both model-driven and data-driven algorithms aim to choose the optimal IKS based on various criteria. In model-driven algorithms, the selection is typically guided by pre-defined conditions such as avoiding singularities or joint limits, choosing configurations closest to a home or zero position, or minimizing the norm of joint angles. These methods are grounded in well-understood mathematical principles and are particularly effective in scenarios where the system's DoF are limited and the kinematic models are well-defined, allowing for precise and reliable solutions.

On the other hand, data-driven methods offer a more flexible approach by considering multiple criteria, such as smoothness, joint limits, operational cost, and energy consumption, to select the optimal IKS. The data-driven models, trained on historical data, are often used to predict and choose the most suitable configuration based on specific constraints or objectives. These

TABLE 7 | Other data-driven methods implemented for IKP of serial manipulators.

Reference	Year	Data-driven methods	Manipulator	DoF	Validation simulation/ experiment	Inference
Duymazlar and Engin (2023)	2023	Boomerang	ABB IRB120	6	Experiment	The noted mean PE and OE were 0.041 mm and 0.076 deg, respectively.
Kumhar and Kukshal (2023)	2023	CNN	—	6	Simulation	The proposed CNN established an effective IK performance.
Singh et al. (2023)	2023	PSO + ANN	—	4	Simulation	The hybrid model reduced the error to below 1.00 mm, surpassing other techniques (> 40.0% improvement).
Kang et al. (2022)	2022	Collision-cost prediction network	KUKA IIWA	6	Experiment	The PE and OE that arrived through the suggested method were significantly better than their counterparts.
Tagliani et al. (2022)	2022	GA + ANN	FANUC LRMate200-iC	6	Simulation and experiment	Validated in both simulation and practical articulated robot, the suggested method significantly reduces MSE.
C. Liu et al. (2022)	2022	IPSO + BPNN	UR3	6	Experiment	Experimental results show effective performance, with joint angle output error below 0.1 deg, surpassing BPNN and PSO + BPNN.
X. Li et al. (2022)	2022	DRL + FFNN	KUKA LBR 7	7	Experiment	The proposed approach confirmed better IK performance than conventional FFNN and DRL.
Mendonca et al. (2022)	2022	ANN	—	2	Simulation	The ANN performed better than the standard GA in providing IKS.
Aggogeri et al. (2022)	2022	GA + ANN	—	3	Simulation	The observed maximum PE was 0.27 mm.
Chandan et al. (2021)	2021	PSO + FA	—	7	Simulation	Catia V5 was used to get position data and by using PSO and FA, required IKS was achieved.
Dong et al. (2021)	2021	RNN + Constrain function	ROBAI CYTON GAMMA 300	7	Simulation	The noted PE was 2.32×10^{-1} mm with a computation time of 4.30×10^{-4} s.
Jiménez-López et al. (2021)	2021	ANN, NR	PUMA	3	Simulation	The ANN exhibited better IK performance than the NR algorithm.
Y. Zhang et al. (2020)	2020	PSO + RBFNN	FANUC 100i	6	Experiment	The accuracy of the IKS has been improved by the recommended method.
Demby's et al. (2019)	2019	ANN, ANFIS	—	4 & 7	Experiment	The experiments (with 4-DoF to 7-DoF robots) suggest that ANN and ANFIS have practical advantages over other algorithms in the literature.

(Continues)

TABLE 7 | (Continued)

Reference	Year	Data-driven methods	Manipulator	DoF	Validation/ simulation/ experiment	Inference
Wu et al. (2019)	2019	PSO + ELM	—	6	Simulation	The simulation results show that PSO + ELM has shown effective performance in solving the IKP of the robot.
D. Zhang and Hannaford (2019)	2019	Behavior tree	PUMA	6	Simulation	Effective inverse IK performance was observed.
Polyzos et al. (2019)	2019	Autoencoder	—	3	Simulation	The recommended algorithm was able to produce IKS successfully.
Y. Zhang and Liu (2018)	2018	IGA + RBFNN	COMAU NJ-220	6	Experiment	The IGA optimizes the network structure and experimental results on a 6 R robot demonstrate the significance of the proposed IKS.
Zeng et al. (2018)	2018	GA + PSO	JACO2 ARM	7	Simulation	Better accuracy and lower computation cost were noted in the developed method.
El-Sherbiny et al. (2018a)	2018	ANN, ANFIS, GA	—	5	Experiment	The experimental test using the 5-DoF robot arm validates that the ANN and ANFIS can be a better choice for proving IKS.
Zhou et al. (2018)	2018	IGA + ELM	STANFORD MT-ARM	6	Experiment	The suggested method significantly reduces computational time and has better accuracy.
Jiang et al. (2017)	2017	PSO + BPNN	UR3	6	Simulation	The noted PE and OE were 0.1 mm and 0.1 deg, respectively.
Starke et al. (2016)	2016	GA + PSO	—	6, 7 & 10	Simulation	Simultaneous exploitation of local extrema enhances accuracy, outperforming model-driven methods in IKS.
Son et al. (2014)	2014	DE + BP	—	4	Simulation	The PE developed by the proposed technique was found to be below 9.0 mm.
B. B. Jha and Biswal (2014)	2014	PSO + MLPNN	PUMA	6	Simulation	The combination of PSO and MLPNN has been noted to exhibit a faster convergence rate and enhanced accuracy with minimal error.
Köker (2013)	2014	ANN + GA	STANFORD	6	Experiment	In multiple studies, PE was noted to be between 0.003 and 4.578 mm.

TABLE 8 | Comparison of the methods.

Methods		Memory requirement	Convergence speed	Interpretability	Robustness	Scalability
Model-driven	Numerical (Hassan et al. 2020; Olsen and Petersen 2011)	High	Medium	Low	Medium	Medium
	Analytical (Singh et al. 2021)	Medium	Medium	Medium	Medium	Low
Model-free/ Data-driven	Algebraic (C. Zhao et al. 2024)	Medium	Medium	Medium	Medium	Low
	Geometric (Selig 1996)	Low	Medium	Medium	Medium	Low
	PSO (Priyadarshi 2024a)	Low	Medium	Low	Low	Low
	GA (M. Wang, Cui, et al. 2018)	High	Slow	Low	High	Low
	ACO (Priyadarshi 2024b)	Low	Medium	Low	Medium	Medium
	SA (Stigler 1981)	Low	Low	Low	Medium	High
	DE (Thakkar Mansi and Patel 2018)	Medium	Medium	Low	Medium	Medium
	Evolutionary strategies (Y. Wang et al. 2007; Bian and Priyadarshi 2024)	Medium	Medium	Low	Medium	Medium
	Machine learning					
	ANN (Haritha et al. 2023)	Medium	Medium	Medium	High	Medium
Hybrid (Haritha et al. 2023; Yadav and Anubhav 2020)	ANFIS (Vargas et al. 2024)	Medium	Medium	High	High	Medium
	RL (Buşoniu et al. 2018; H. Liu et al. 2023)	High	Medium	Medium	High	High
		High	Medium	Medium	High	High

data-driven techniques are advantageous in complex, high-dimensional, or uncertain environments where model-driven methods may fall short. They excel in exploring diverse solution spaces, adapting to intricate or multi-modal IKP, and enhancing real-time performance by managing multiple configurations effectively.

5.1 | Limitations and Challenges

The intricate nonlinear relationship between joint variables, end-effector orientation, and position, solving the IKP for manipulators proves to be quite complex. Moreover, the singularity issue is considered one of the greatest hurdles while handling IK.

In the case of model-driven methods, they have a huge limitation while dealing with highly nonlinear, multi-DoF, and highly redundant robotic manipulators. The desired solutions are not reached while handling such situations. Furthermore, these methods require a deeper mathematical knowledge is very much required to attempt and successfully execute the IK of the manipulator. This is one of the main challenges of model-driven methods. Wherein, numerical-based IKS encounters complexity hurdles, particularly when dealing with redundant manipulators. The geometric method hinges on obtaining closed-form solutions for the initial three joints within a geometric context, and their effectiveness is dependent on the manipulator's structure and DoF. While suitable for systems with ≤ 3 DoF, it necessitates various approximations and simplifications for higher DoF scenarios. The analytical method (iterative), though inherently time-consuming, may yield redundant or inadequate solutions, especially in the presence of singularities in the robot's geometric model. Additionally, the accuracy of the iterative technique often relies on early iterations. Followed by algebraic-based IKS gives less accuracy while dealing with high DoF and it relies on system configuration and FK modeling. Systems with singularities in their forward model may not be compatible with algebraic solutions.

The primary limitations of data-driven methods include potential overtraining, substantial input requirements, challenges associated with network optimization, and errors in extrapolation. Moreover, the effectiveness of data-driven approaches is intricately tied to parameter selection, requiring careful tuning for optimal performance. Data-driven methods, despite their flexibility, come with certain constraints when applied to solving IKP. One notable challenge is the potential for suboptimal convergence, as these methods may not consistently yield the most efficient or accurate solutions. Especially in supervised ANN-based IKS, another obstacle arises with the pre-processing and post-processing of data. Handling data pre-processing and error post-processing is a significant challenge, particularly due to the vast volume of data involved. The immense data sets often result in issues with noise and imprecise information, making them challenging to address. Data pre-processing typically involves tasks such as data augmentation, correlation analysis, data clustering, normalization, resolution adjustment, graph construction, and filtering out faulty and missing data. Meanwhile, post-processing steps may encompass various rule filtering or the integration of knowledge and pruning techniques. Still, these algorithms require high-end



FIGURE 16 | Challenges in optimization of ML algorithms. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70014)]

computers to perform training and testing which seems to be an expensive way. Offline IK approaches struggle to handle unpredictable conditions as they occur. Implementing online data-driven IK methods enables the attainment of nearly optimal IKS and enhances system stability. However, this necessitates adequate storage capacity and frequent updates to the model. Furthermore, hybrid data-driven-based IKS shows significant results in solving IKP, leading to a growing number of papers exploring their hybridization. Generally, in hybrid data-driven methods, metaheuristic algorithms are used to optimize the ML algorithms. The hybrid data-driven models emphasize the importance of balancing high accuracy with manageable complexity. The challenges involved in optimizing the ML algorithms are given in Figure 16.

5.2 | Future Research Direction

With advancements in artificial intelligence, ML, and computational capabilities, the potential for enhancing the efficiency and versatility of robotic manipulators is vast. In this context, exploring avenues to refine and innovate IK techniques becomes paramount. A few such avenues are mentioned below.

- An improved FABRIK method has more scope to enhance the IKS of the manipulator. Thus, researchers can analyze the methods and ways to explore the different streams in improving the FABRIK method without increasing the computational burden. Moreover, the enhanced hybrid model-driven approach also has a better future scope.
- Integrating online learning techniques into data-driven models for IKP enables robots to continuously adapt to evolving scenarios without requiring extensive retraining. Online learning, a paradigm in ML, updates models incrementally (self-tuning) with new data, making it suitable

for applications demanding adaptation to changing conditions over time.

- Employing data-driven methods enhances human-robot interaction by enabling robots to understand and adapt to human motions and preferences in collaborative tasks. Exploring transfer learning methods in robotic systems aims to boost adaptability by leveraging knowledge gained from solving IKP in one scenario and applying it to related contexts. Transfer learning improves learning efficiency, reduces the need for extensive data sets, less computation burden, enhances generalization, and overall improves IK performance.
- Researchers can explore data-driven algorithm optimization techniques through graph data. In this way, it will facilitate effective inference and learning of the data-driven algorithm. Consequently, optimization methodologies, such as novel evolutionary algorithms inspired by biological and social systems may be attempted to effectively optimize the data-driven approach. In addition, researchers can explore neuro-symbolic optimization techniques. This field explores the amalgamation of ANNs and symbolic reasoning in optimization protocols, leveraging the merits of each paradigm to enhance overall performance.
- Another important area to explore data-driven methods is optimizing under resource-constrained environments. Formulating optimization strategies that optimize model performance under limitations such as processing, energy, and memory limitations has a huge future scope.
- Applying chaos theory to IKP involves studying the sensitivity of robotic systems to initial conditions, leading to innovative solutions for achieving desired end-effector positions.
- Quantum-based algorithms hold the potential to efficiently address complex mathematical problems, offering accelerated solutions to high-dimensional IKP.
- Holographic computing, which utilizes holograms to encode and process information, has the potential to be integrated into IK, offering distinctive representations of robot configurations that enhance visualization and control capabilities.
- By integrating both model-driven and data-driven approaches, the algorithm can leverage the robustness of established methods while exploiting the flexibility and learning capabilities of modern techniques. This combination ensures more efficient and accurate handling of the various joint space configurations, leading to better overall performance in solving IKP.

The primary focus of all the above-mentioned research perspectives will be refining algorithms and techniques to empower manipulators to navigate complex environments with increased precision. Moreover, the future advancement of IK will rely heavily on data-driven algorithms with reduced computational burden. The compressed data-driven methods may be a potential solution. Most IK methods have primarily been validated through simulation. However, minimal effort has been dedicated to real-world validation, resulting in a significant decrease in performance accuracy when applied to actual manipulators. Therefore, researchers can explore giving more preference to validate the IK methods in real-time.

6 | Conclusion

Serial manipulators find extensive application across diverse industries, contributing to reduced human effort, enhanced safety, and improved production efficiency. Nevertheless, the solution to the IK problem for serial manipulators is recognized as challenging, time-consuming, and inherently nonlinear. Addressing these widespread research problems, such as reducing computation time, obtaining accurate results, and optimizing path planning, becomes of paramount importance. This article delves into a comprehensive exploration of the IK methods and future prerequisites for the continuous advancement of diverse methods in solving the IK of serial manipulators. It identifies substantial limitations and challenges necessitating resolution and presents state-of-the-art techniques meticulously designed to enhance serial manipulator performance. The primary objective is to provide an overview of crucial methods utilized in the IK of serial manipulators over the past decade (2014–2023). A systematic protocol was adopted and by using appropriate keywords, about 321 articles were gathered. Then, the IK methods were broadly classified as model-driven and model-free/data-driven methods. Some conclusions have been drawn regarding each technique and they are given below.

Under model-driven methods, discussions cover analytical, algebraic, numerical, geometric, and other model-driven methods, revealing drawbacks in terms of robustness and efficiency. Thus, relatively few publications are noted in this category. However, the performance of hybrid model-driven algorithms and FABRIK method was relatively better than other model-driven methods. Improved FABRIK method and enhanced hybrid model-driven method can be future research directions in the case of model-driven IK methods. Conversely, data-driven methods prove more suitable for IK, offering adaptiveness that learns from mistakes and improves in subsequent occurrences. Hence, more publications are observed in data-driven-based IK methods. The model-driven methods includes: metaheuristic (particle swarm optimization, genetic algorithms, and other metaheuristic methods), machine learning (artificial neural networks, neuro-fuzzy, and reinforcement learning), and other data-driven methods. These methods show relatively better performance compared to model-driven techniques. Having said that, these methods have a higher computational burden and mostly rely on the supplied data set. As a future research direction, data-driven algorithms may have the least computational burden and can be more tolerant to resource-constrained environments (limited processing, energy, memory, and data set).

In an era marked by advanced technology and widespread information, researchers continue to devise novel methods to address the IK of serial manipulators, garnering significant research interest. The findings in this paper offer valuable insights for researchers and robotics industries, serving as a reference guide and also helping in identifying areas for new research directions in the field of IK of serial manipulators.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no data sets were generated or analyzed during the current study.

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